

European Network of
Transmission System Operators
for Electricity

# ENTSO-E

## CGM BUILDING PROCESS IMPLEMENTATION GUIDE

## AC PART

FOR SYSTEM OPERATIONS

## 1.3 EDITION

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COMMON GRID MODEL BUILDING PROCESS (CGM_BP)

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# NOTE CONCERNING WORDING USED IN THIS DOCUMENT

The force of the following words is modified by the requirement level of the document in which they are used.

MUST: This word, or the terms “REQUIRED” or “SHALL”, means that the definition is an absolute requirement of the specification.

MUST NOT: This phrase, or the phrase “SHALL NOT”, means that the definition is an absolute prohibition of the specification.

SHOULD: This word, or the adjective “RECOMMENDED”, means that there may exist valid reasons circumstances to ignore a item, but the full implications must be understood and carefully weighed before choosing a different course.

SHOULD NOT: This phrase, or the phrase “NOT RECOMMENDED”, means that there may exist circumstances when the particular behaviour is acceptable or even useful, but the full implications should be understood, and the case carefully weighed before implementing any behaviour described with this label.

MAY: This word, or the adjective “OPTIONAL”, means that an item is truly optional. One vendor may choose to include the item because a marketplace requires it or because the vendor feels that it enhances the product while another vendor may omit the same item. An implementation which does not include an OPTION MUST be prepared to interoperate with another implementation which does include the OPTION, though perhaps with reduced functionality. An implementation which does include an OPTION MUST be prepared to interoperate with another implementation which does not include the OPTION (except, of course, for the feature the option provides.)

DEPRECATED: this word means that a previously permitted entity should no longer be used in new implementations as in a future release the object in question may be suppressed.



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# 1. Introduction

This implementation guide advises TSOs how to model electric power systems, so data can be exchanged and used by the merging agent to perform the merging process, which consist of assembling Individual Grid Models (IGM) into a Common Grid Model (CGM). The merging process is performed repeatedly for different timeframes to serve the different RSC services that are using the CGM: Coordinated Capacity Calculation, Coordinated Security Analysis, Outage Planning and Short-Term Adequacy Forecast.

The CGM creation process has two-steps. In the first step TSOs, as a modelling authority, create power system models covering at least the network regions they are responsible of, and perform power flow calculations for it. An IGM is extracted from the power system model and power flow calculation result. In the second step merging agents (e.g. Regional Security Coordinators – RSCs) collect and assemble IGMs into merged models for the synchronous area where an RSC is operating up to pan-European extent. As of Article 2(2) of Regulation (EU) 2015/1222 the term CGM refers to a Union-wide data set agreed between various TSOs describing the main characteristic of the power system (generation, loads and grid topology) and rules for changing these characteristics during the capacity calculation process. There can be similar models on the level of synchronous areas which strictly are no CGMs (not Union-wide) but merged models. In the document, whenever the term CGM is used it stands for both a CGM and a merged model except where the merged model is explicitly mentioned.

The CGM creation is envisaged as a fully automated process that does not require human attention if the power flow results satisfy the criteria stipulated for the RSC services.

The exchange of IGMs and CGMs is based on the IEC TS 61970-600-1/2:2017 technical specification ([2] in section 3). IEC TS 61970-600-1/2:2017 have profiles that divide data in groups with different exchange cycles, e.g.

- EQ profile, an equipment instance file that describes the equipment in the power system model covered by a Modelling Authority Set, is updated at least when the equipment is changed. This data can be exchanged when it changes or at a fixed cycle with highest resolution of exchange for every market time unit, following the established processes and restrictions imposed by the storage and archiving rules of the applications connected to the ENTSO-E operational planning data environment (OPDE).
- SSH profile, a steady state hypothesis instance file that contains all objects required to exchange input parameters to be able to perform load flow simulations, exchanged for every market time unit.
- TP profile, a topology instance file that contains all topology objects for a Modelling Authority Set referencing the corresponding equipment and describing how equipment is electrically connected, exchanged for every market time unit.
- SV profile, a state variable instance file that contains all objects required to complete the specification of a steady-state power flow solution, exchanged for every market time unit.

A typical scenario would be to exchange one EQ profile and one set of SSH, TP and SV profiles for several time steps, e.g. for every market time unit, depending on a time horizon. E.g. in



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case of day-ahead congestion forecast process 24[1] sets are covering an energy delivery day but in case of intraday the data exchange can start up to 29[2] hours ahead, see also Table 3 and Table 4 further down in this document.

For a power flow calculation to create meaningful and reliable result, the exchanged data must meet quality criterions specified in detail in the documents IEC TS 61970-600-1/2:2017 [2] and QoCDC [3], i.e.

- The equipment data must be accurate and with good fidelity describe the real equipment. If this is not the case the power flow result may be meaningless.
- The market schedules must be mapped to power flow inputs close to loads and productions as they appear at injection points on the borders of the IGM. If this is not the case solved power flows and area interchanges may not be representative.
- Modelling of controls and constraints need to reflect real behaviour and limit values so power flow calculation response to contingencies is accurate.

The CGMES allows for detailed modelling of power networks, traditional bus branch style models as well as node breaker models including measurements. The detailed modelling enables better alignment between power flow results and the real state in the modelled power system. CGMES also allows detailed models to be mixed with bus branch style models which enables a gradual transition to more detailed modelling. The capability for detailed modelling also means more data to manage, keep consistent and of good quality.

# 2. Scope

## 2.1 Concept of the document

This document contains not only guides for implementation but also other information that support implementation of the CGM building process.

In the future this document will be superseded by standards that cover the content in the different sections of the document. When this happens, this document will be shaped to implementation guide to suit those standards, as complex processes always require guidelines.

## 2.2 Who should read what?

This document provides guides to various parts and aspects of the CGM building process. Extensive information about all cases, experiences, and relevant details gathered during ten years of interoperability testing is provided. In order to focus different readers on the information, which is relevant to their area of responsibilities, here below is listed breakdown of chapters.

All readers should gain the basic understanding of the CGM building process based on the following chapters:

**3 Related Documents**

[1] 23 (spring) and 25 (autumn) daylight saving

[2] 30 in case of autumn daylight saving

**4 Definitions**

**5 Overview of the rule levels**

The purpose of this section is to provide an understanding of the type of data validated at each level. This is an informative section with guiding information to provide a short link to QoCDC [3] document.

**6 Business process**

This section describes the business process including the actors, the exchange of CIMXML documents, validation of IGMs and CGMs, creation and validation of IGMs, assembly and validation of CGMs and power flow calculations for capacity/congestion evaluation.

Please also refer to the “EMF requirement specification” section 3 [4] that provide more details.

**7 XML format**

This section contains guiding information, descriptions and restrictions on the XML format.

Harmonisation of current and revised meta data information model is described in chapter:

**8 Metadata**

The combination of the normative and guiding information on meta data rules is provided in this chapter and its subchapters. This chapter depend on QoCDC section 3 [3], describing the current and partly implicit meta data information model as well as the revised information model.

Network system modelers and power system analysis engineers will find details on refinement of modelling of power system equipment in chapters:

**9 Boundary data**

- This section is mainly guiding but has statements of normative nature.

**10 Network data**

This section discusses modelling issues discovered during analysis of IGMs and interoperability tests. The issues should be feedback to standardisation work so that the standards can be improved. Some issues can be solved by adopting one of several possible modelling styles. This is a guiding section.

## 3. Related Documents

The following documents are applicable:

[1] The IEC TS 61970-600-1/2:2017 is based on the Common Information Model (CIM) 16 (UML 16v29). The following documents defines the semantic model:

- IEC 61970-301:2016 Ed6: Common Grid Model (CIM) Base
- IEC 61970-302:2018 Ed1: Common Grid Model for dynamics specification
- IEC 61970-452:2017 Ed3: CIM Static Transmission Network Model Profiles (i.e. the



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network equipment model)

- IEC 61970-456:2018 Ed2: Solved Power System State profiles
- IEC 61970-453:2014 Ed2: Diagram Layout profile
- IEC 61970-552:2013 Ed1: CIM XML Model Exchange Format
- IEC 61970-501:2006 Ed1: Common Information Model Resource Description Framework (CIM RDF) schema

[2] More restricting is the Common Grid Model Exchange Specification version 2.4.15 (CGMES2.4.15) now published as the specification IEC TS 61970-600-1/2:2017, providing context for ENTSO-E exchanges:

- IEC TS 61970-600-1:2017 Ed1: Common Grid Model Exchange Specification
- IEC TS 61970-600-2:2017 Ed1: Exchange profiles specification

[3] QoCDCv3.2 The validation rules are specified in the “Quality of CGMES Datasets and Calculations” (QoCDC) document, which also includes the file name convention for IGM files.

[4] The merging process, solving power flow on merged models (CGMs) and verify that the power flow solution is valid and safe is specified in the “EMF requirement specification” document. The document specifies requirements needed to enable a highly automated exchange process and univocal interpretation of exchanged network models for the CGM process using IEC TS 61970-600-1/2:2017.

[5] CGMA and PEVF specifications describe the schedule data used to create the inputs for solving power flows at every market time unit. The documents can be found at the ENTSO-E EDI library (https://www.entsoe.eu/publications/electronic-data-interchange-edi-library/ ).

# 4. Definitions and abbreviations



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| Term | Description |
|---|---|
| Assembled model | An IGM is assembled by pulling together CIMXML files tracing by the md:Model.DependentOn references starting from the SV CIMXML file.<br>A CGM is assembled by pulling together the CIMXML files for the IGMs to be included in the CGM. |
| Boundary set | A boundary set consist of the CIMXML files EQBD and TPBD. |
| Market time unit | This is the term used for the schedule time steps in the documents CGMM and CACM. Same as effectiveDateTime and Model.scenarioTime in |
| Bus Branch (BB) model | A network model using a simplified representation of a network. Detailed breaker bays are normally not described by cim:Switches and if cim:Switches are used they shall be flagged as retained. |
| Node Breaker (NB) model | A network model representing breaker bays with cim:Switches flagged as non-retained. Several levels of details are possible. The simplest is just describing breakers with cim:Breakers and the more detailed is also describing isolators and disconnectors with cim:Disconnectors. Some cim:Switches may be flagged as retained in case the power flow across them need to be monitored, e.g. coupler bays. |
| TN | TopologicalNode represents the power flow busses in a power flow model, defined as the single fictive point in a system at which the several components of the power system like generators, loads, and feeders, etc., are connected; hence, fictive power flow busses, associated with four quantities of a load flow algorithm: voltage magnitude, phase angle of the voltage, active power, reactive power. |

313

| Term | Description |
|---|---|
| CN | ConnectivityNode represents the electrical connection points in a network model, is always used in NB models and could be used in BB models.<br>Note that NB models may have parts that are BB style where detailed switch modelling has not been used. |
| IGM | Individual Grid Model. A network model provided by a Modelling Authority (e.g. a TSO). An IGM is a collection of CIMXML files for a market time unit of energy delivery day and a time horizon, e.g. one day ahead. |
| CGM | Common Grid Model. A power system model provided by a Merging Agent (e.g. an RSC). A CGM is assembled from IGMs for a market time unit of energy delivery day and time horizon, e.g. one day ahead. |
| CGMM | Common Grid Model Methodology. |
| CACM | Capacity Allocation and Congestion Management |
| RSC | Regional Security Coordinator |
| RCC | Reactive Capability Curve |
| OPDM | OPERATIONAL PLANNING DATA MANAGEMENT is a file exchange service for remote clients. It is used to exchange CIMXML files as well as other files. |
| QAS | Quality Assurance Service |
| NMD | Network Modelling Database |

## 5. Overview of the rule levels

The QoCDC document ([3] in section 3) define rules that are divided in levels where the lower level rules must be passed before higher-level rules can be processed. The rules are organized in eight levels with complexity increasing at higher levels as listed in Table 1.

320 **Table 1 Validation levels**

| # | Level name | Level description |
|---|---|---|
| 1 | Meta data in file names | Level 1 checks CIMXML file naming, meta data in file names and file packaging. This level supports the automated file exchange and processing. |
| 2 | Structure syntax and metadata | Level 2 checks XML and RDFS validity, meta data in CIMXML file header and consistency with meta data in CIMXML file name. This level supports the automated file exchange and processing. Level 2 contains rules checking different types of data<br>- XML format issues.<br>- Meta data issues.<br>- Network model profile related issues. |
| 3 | Constraints and mapping | Level 3 checks the validity of objects in the scope of a single CIMXML file. |
| 4 | Model assembly | Level 4 checks that the meta data describing dependencies between CIMXML is consistent so that the files can be assembled. This level supports the automated file exchange and processing. Level 4 contains rules checking different types of data<br>- Meta data issues.<br>- Network model profile related issues. |
| 5 | Consistency of assembled model | Level 5 checks that objects across CIMXML files for different profiles in an assembled IGM or CGM are consistent. |
| 6 | IGM and CGM plausibility | Level 6 checks that the power flow solution provide a good enough input for subsequent CGM assembly. |
| 7 | Coordination | Level 7 checks that the assembled IGMs in a CGM are consistent with each other as well as with market data (PEVF and CGMA files). |
| 8 | Convergence behaviour and CGM plausibility | Level 8 checks that IGM or CGM power flow convergence is well behaved and that a solution is plausible, e.g. stays within given limits. |

321

322 Table 1 shows the validation levels where the lower levels discover fundamental issues that
323 blocks further processing. For CIMXML files to be useful they must pass levels 1 to 6. This is
324 elaborated in section 5.

# 6. Business process

## 6.1 Introduction

This section gives an overview of the assembly process and guides on how to specify meta data controlling the process and the exchange of data.

The roles appearing in the CGM building process are shown in Figure 1.

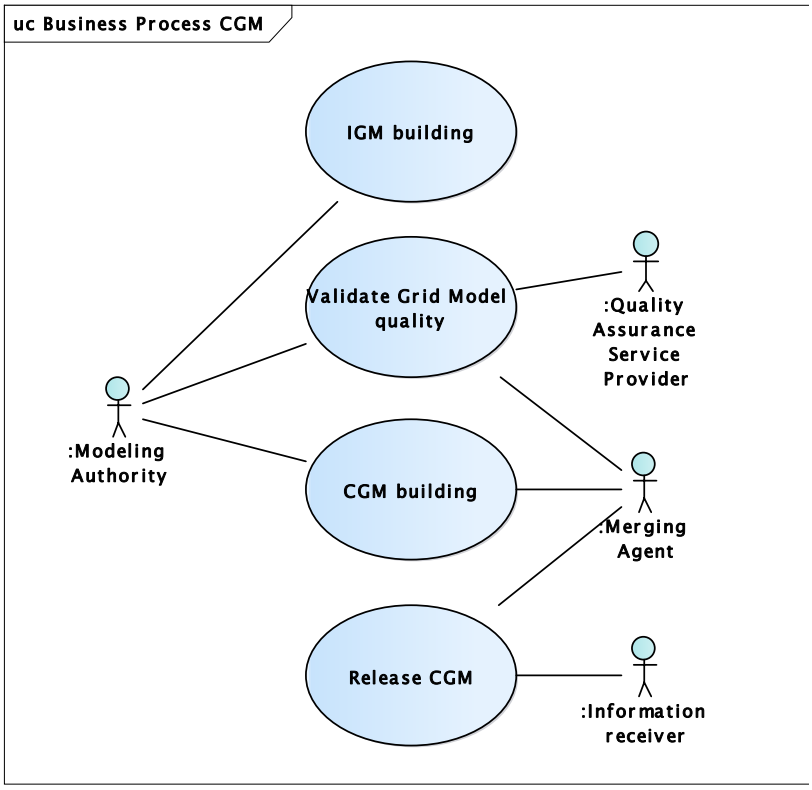



**Figure 1 Use case diagram of the CGM building process**

The roles involved in the CGM building process are:

- Alignment Agent that provide schedules for the respective energy delivery day, not shown in Figure 1.
- Modelling Authorities (TSOs) that use the schedules to evaluate network security or transfer capacity for each time unit. This result in IGMs made available to other actors after validation by a Quality Assurance Service Provider.
- The Quality Assurance Service Provider validates IGMs and CGMs.
- Merging agents (e.g. RSCs) are responsible for assembling IGMs into a CGM which is made available to Information Receivers after validated by a Quality Assurance Service Provider.
- Information Receivers are users of validated IGMs or CGMs. The use is typically to perform an analytical study with the IGMs or CGMs as base. Examples of Information Receivers are TSOs, RSCs, ACER etc.

The interaction between the Alignment Agents and the Modelling Authorities (TSOs) is not covered by this document but the interaction between Modelling Authorities (TSOs) and

Merging agents (RSCs) is described. The steps in the interaction are shown in Figure 2 and Table 2.

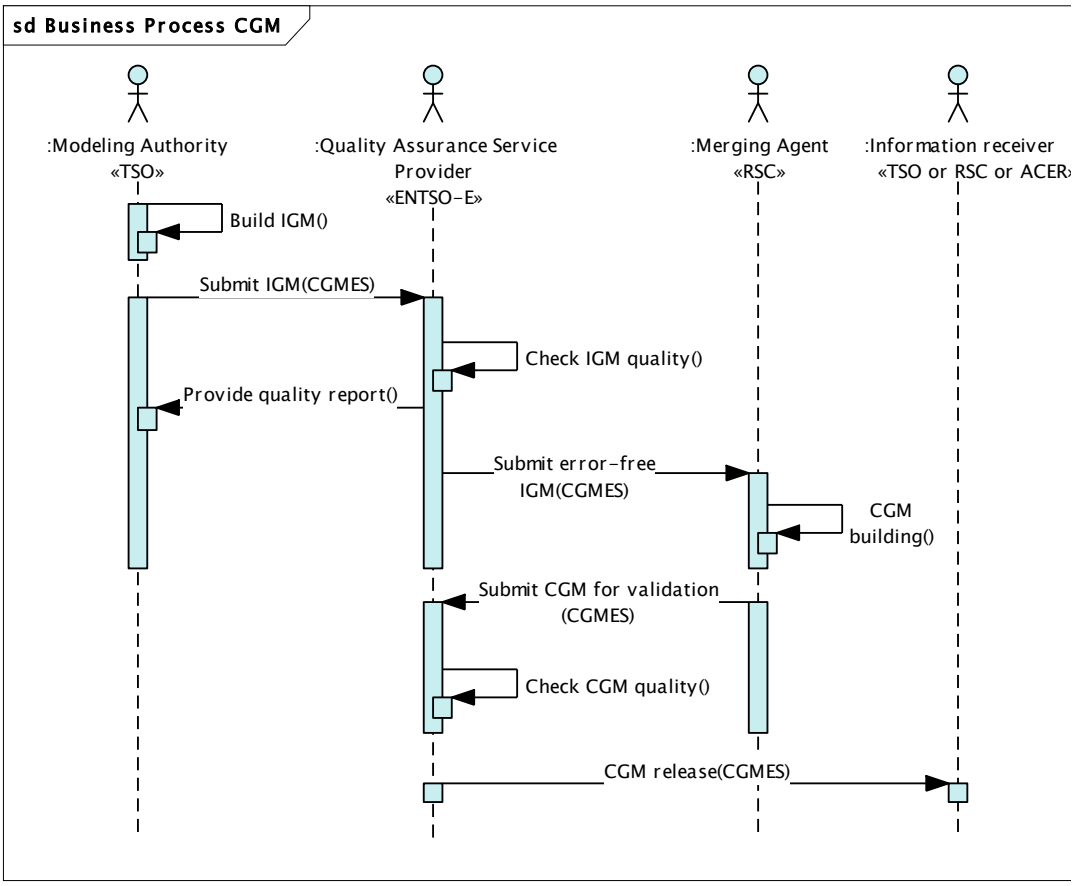



**Figure 2 High level sequence diagram of the CGM building process**

The power flow cases prepared in steps 2 and 7 require that the network models have a well-defined boundary set that separates the IGMs. IGMs are non-overlapping and the boundary establishes electrical connection points (also called X-nodes based on UCTE area code legacy) between the IGMs, e.g. where tie lines connect. The boundary is described by two CIMXML files:

- EQBD profile - a boundary equipment instance file that contains all objects defined in the CGMES boundary equipment profile and includes data for boundary information relating to a given exchange
- TPBD profile - a boundary topology instance file that contains all objects defined in the CGMES boundary topology profile and includes data for boundary information relating to a given exchange

Boundary equipment (EQBD) and boundary topology (TPBD) instance files, referred together as boundary set (BDS) contain data needed to create an individual grid model (IGM) as well as to create a pan-European or regional model. The BDS is currently maintained centrally by ENTSO-E in the Network Modelling Database (NMD). As an interim solution, the duty of the BDS provision to OPDE has been delegated to one of the OPDE Operators, consisting of the monthly upgrade of the BDS provided by NMD in fit-for-purpose form, based on current

368 requirements, making it available on OPDE via a dedicated OPDM client and tagging it as
369 official.

370 The steps in Table 2 are further detailed in the following sections.
371

372 **Table 2 Processing steps in tabular form**

| # | Step description |
|---|---|
| 1 | The Modelling Authority (TSO) receives market schedules on daily basis (process description is outside the scope of this document). |
| 2 | Modelling Authority (TSO) creates a power system model of its area of responsibility with a solved power flow case for each market time unit. |
| 3 | Modelling Authority (TSO) creates a set of CIMXML files for the solved power flow cases, an IGM that contains of:<br>a) Equipment file (EQ) per region (if applicable), per business process (if relevant) with provision frequency spanning from “when change occurs” up to per market time unit.<br>b) Topological processing and power flow solution and results for each market time unit which means that the solution appears in the following CIMXML files<br>i. One SSH CIMXML file containing the power flow inputs, the solution and switch statuses if present.<br>ii. One TP CIMXML file containing topology objects (fictive power flow busses) referencing the corresponding equipment and defining how the equipment is electrically connected.<br>iii. One SV CIMXML file that contains the same solution data as the SSH CIMXML file, see also section 6.4. |
| 4 | A Modelling Authority (TSO) submits CIMXML for validation to a Quality Assurance Service Provider. |
| 5 | A Modelling Authority (TSO) CIMXML error-free in validation levels 1-6 files are made available to Merging Agent (RSCs). If errors are detected the IGM is rejected. |
| 6 | A Merging Agent (RSC) obtains the IGM CIMXML files for Modelling Authorities (TSOs) within a region of concern. |
| 7 | The Merging Agent (RSC) creates the CGM and power flow is solved for each time unit. |
| 8 | The Merging Agent (RSC) provides the updated power flow inputs (SSHrep, for more details see section 6.6) per Modelling Authorities (TSO) and results per region (SV) for CGM for validation to Quality Assurance Service Provider. If error-free, CGM is made available to Information receiver. If errors during validation are detected, the CGM is rejected. |

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## 6.2 Overview CIMXML files

The data exchanged in the workflow steps described in Table 2 is primarily CIMXML files for IGMs and CGMs. The CIMXML files for an IGM are listed in Table 3.

**Table 3 CIMXML files in an IGM**

| Type of CIMXML file | Appearance |
|---|---|
| EQBD<br>Boundary Equipment | One for the Pan-European grid model. |
| TPBD<br>Boundary Topology | One for the Pan-European grid model. |
| EQ<br>Equipment, described by three different EQ profiles:<br>- EQ Core - all equipment<br>- EQ Operation - data needed by real time systems, e.g. SCADA/EMSes<br>- EQ Short circuit - data needed by short circuit calculations | One per TSO (and region if applicable).<br>Per business process (e.g. 1D, 2D) or without indication of business process if relevant for all.<br>This data can be exchanged:<br>- at least when it changes, following the established processes and restrictions imposed by the storage and archiving rules of the applications connected to the ENTSO-E operational planning data environment (OPDE)<br>- at a fixed cycle with highest resolution of exchange for every market time unit |
| SSH<br>Power flow input | One per energy delivery day, business process, market time unit and TSO. |
| TP<br>Fictive power flow busses | One per energy delivery day, business process, market time unit and TSO. |
| SV<br>Power flow solution | One per energy delivery day, business process, market time unit and TSO. |

A Common Grid Model (CGM) is established/assembled as a result of CGM Building process in line with legal requirements arising from EU Network codes.

A CGM consists of several assembled IGMs and the CIMXML files describing a CGM are listed in Table 4.



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385 **Table 4 CIMXML files in Common Grid Model (CGM)**

| Type of CIMXML file | Source | Appearance |
|---|---|---|
| EQBD<br>Boundary Equipment | BDS | One for the Pan-European grid model. |
| TPBD<br>Boundary Topology | BDS | One for the Pan-European grid model. |
| EQ<br>Equipment | IGM | One per TSO (and region if applicable).<br>Per business process (e.g. 1D, 2D) or without indication of business process if relevant for all.<br>This data can be exchanged:<br>- at least when it changes, following the established processes and restrictions imposed by the storage and archiving rules of the applications connected to the ENTSO-E operational planning data environment (OPDE)<br>- at a fixed cycle with highest resolution of exchange for every market time unit |
| SSH<br>Power flow input | IGM | One per energy delivery day, business process, market time unit and TSO.<br>These files are from the original TSO's IGMs and are unchanged. |
| Updated SSH<br>Power flow input (SSHrep) | CGM | One per energy delivery day, business process, market time unit and TSO.<br>These files are updated SSH CIMXML files of TSO's IGMs and are specific to the CGM, for more details see section 6.6. |
| TP<br>Fictive power flow busses | IGM | One per energy delivery day, business process, market time unit and TSO.<br>These files are from the original TSO's IGMs and are unchanged, no CGM specific TP file exists. |
| SV<br>Power flow solution | CGM | One per energy delivery day, business process, market time unit, RSC and region (synchronous area or Pan-European). |

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387 ## 6.3 Data validity

388 Validity of CIMXML files and assembled models is checked by the Quality Assurance Service.
389 Any reported error at levels 1 to 6 prevents the files from becoming available for further

processing by Merging Agent.

The validity of data is checked against the IEC TS 61970-600-1/2:2017 profile information model ([2] in section 3) and rules described in the QoCDC document ([3] in section 3). The IEC TS 61970-600-1/2:2017 profile is used to validate that exchanged data conforms to the class, attribute and role names defined in the IEC TS 61970-600-1/2:2017 profile and that cardinalities are respected. The rules in the QoCDC document ([3] in section 3)describe additional data and restrictions to IEC TS 61970-600-1/2:2017 necessary to obtain the CIMXML file exchange process operational and the power flow calculations resulting in meaningful solutions.

The levels in Table 1 represent a process where the lower levels must be successfully completed before it is meaningful to proceed with next level. The process description seen from the input data side and necessary input data in certain validation levels is illustrated in Figure 3.

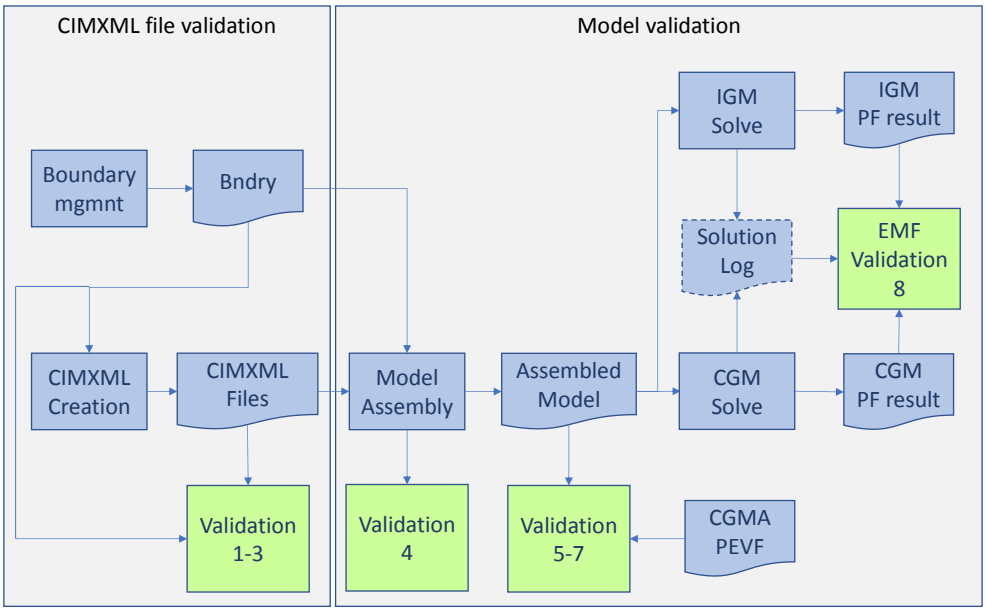



**Figure 3 Validation levels concept for IGM and CGM[3]**

The symbols in Figure 3 has the following meanings

- Blue box – data processing (concept).
- Blue document – CIMXML file(s) or reporting market document file.
- Green box – validation.

The shaded backgrounds “CIMXML file validation” and “Model validation” show the scope for the validation per file and per model (IGM, CGM).

“CIMXML file validation” is done per CIMXML file which means that the information in other CIMXML files is not used. This validation is done on simple data as name lengths but for EQ

[3] Bndry (as Boundary Set), CGMA and PEVF (as reporting market documents) are considered as already validated input data



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files more complex validation is done, e.g. attribute value ranges, curve data and limit values.

For “Model validation” all CIMXML files for an IGM or CGM are assembled into a model before validation is done. A complete model is assembled and validated for each market time unit. Model validation done for IGMs are not applicable for a CGM as the model is a new assembly with updated data. Hence model validation shall be repeated for a CGM. The CIMXML file validation (levels 1, 2 and 3) once done for an IGM can be trusted and need not be repeated for a CGM. A CGM contain new SSHrep CIMXML files for each IGM (see section 6.6 for details) and one new SV CIMXML file for the solution. These files are validated at levels 1-3. The CGM model is validated at level 4 and up.

Errors at levels 1 to 6 will prevent an IGM to become available to a Merging Agent for further processing and creation of CGM, as well as for a CGM to become available to Information receiver (see Figure 2).

The “Boundary mgmnt” box represents the boundary files creation and management (irrelevant to implementation). Hence the “Bndry” document represents the Boundary Equipment and Topology CIMXML files that are also validated at levels 1 to 4 but in Figure 3 they are considered as already validated input data. The “CIMXML creation” box is where a TSO create the CIMXML files, or Merging agent creates CGM CIM XML files (refer to Table 3 and Table 4). The result is the “CIMXML files” that are validated with level 1 to 3 rules.

When an IGM or CGM is assembled it is checked that all necessary CIMXML files are available. This is done in “ModelAssembly” that checks the availability of CIMXML files according to level 4 rules.

The assembly results in an “Assembled Model” that is checked with level 5 to 7 rules. Level 7 rules that check the consistency between IGMs and consistency with the net positions in PEVF or CGMA documents.

Level 8 rules check the power flow result is plausible, the “IGM Solve” path corresponds to the process described in section 6.5 and the “CGM Solve” path corresponds to section 6.6.

## 6.4 IGM creation

A power flow solution described by CIMXML files use the files depending of the profile as follows:

- EQ is an input to power flow describing the network
- SSH describes the power flow input parameters, e.g. injections and set point values.
- TP describes the fictive power flow busses and depends on the type of model
  - For Node Breaker (NB) model’s TP is an output from topology processing, Figure 28 show an architecture for this. TP is an input to power flow.
  - For Bus Branch (BB) model’s TP is manually maintained in tools dedicated to this. TP is an input to power flow.
- SV describes the power flow solution, so it is an output from power flow.

The European network is interconnected, especially central Europe, which means that the impact from the neighbouring networks and market schedules need to be considered.

Creating an IGM for a TSO, means including parts of the neighbouring networks and market schedules. Hence, the power flow will usually be solved not only for the part of power system

that is the area of the responsibility of a TSO but for a wider system of interest and impact. This means that the power system for the IGM needs to be extracted from the wider power system model that is used when solving the power flow for a specific scenario (time horizon, market time unit).

The boundary between the IGMs is described by boundary cim:TopologicalNodes (TN) and boundary cim:ConnectivityNodes (CN), also referred to as boundary points or as X-nodes as UCTE area code legacy. This means that the extraction of an IGM stops at the boundary nodes. Figure 4 shows a boundary as described by CIM/CGMES objects.

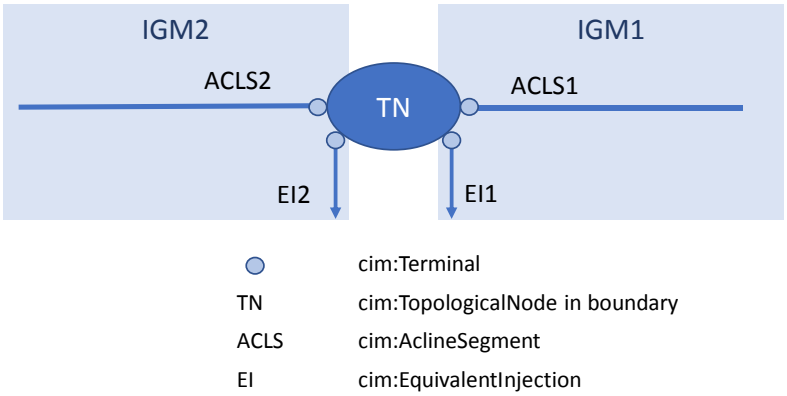



**Figure 4 A boundary in CIMXML**

Figure 4 shows how two IGMs connect at a boundary TopologicalNode (TN). Each IGM connects with one cim:ACLineSegment (a retained cim:Switch or cim:PowerTransformer is also allowed) and one cim:EquivalentInjection. When the IGM is extracted, the power flow across the boundary must be transferred as an injection value to the cim:EquivalentInjection (cim:EquivalentInjection.p and cim:EquivalentInjection.q attributes). The sum of the boundary cim:EquivalentInjection injections (of the tie lines for which a cim:TieFlow exists, for more details see section 9) shall then match up with the net interchange for the IGM (cim:ControlArea.netInterchange attribute, the cim:ControlArea is not shown in the figure).

Figure 5 gives a conceptual overview of the data flows and functions involved in the creation of an IGM.

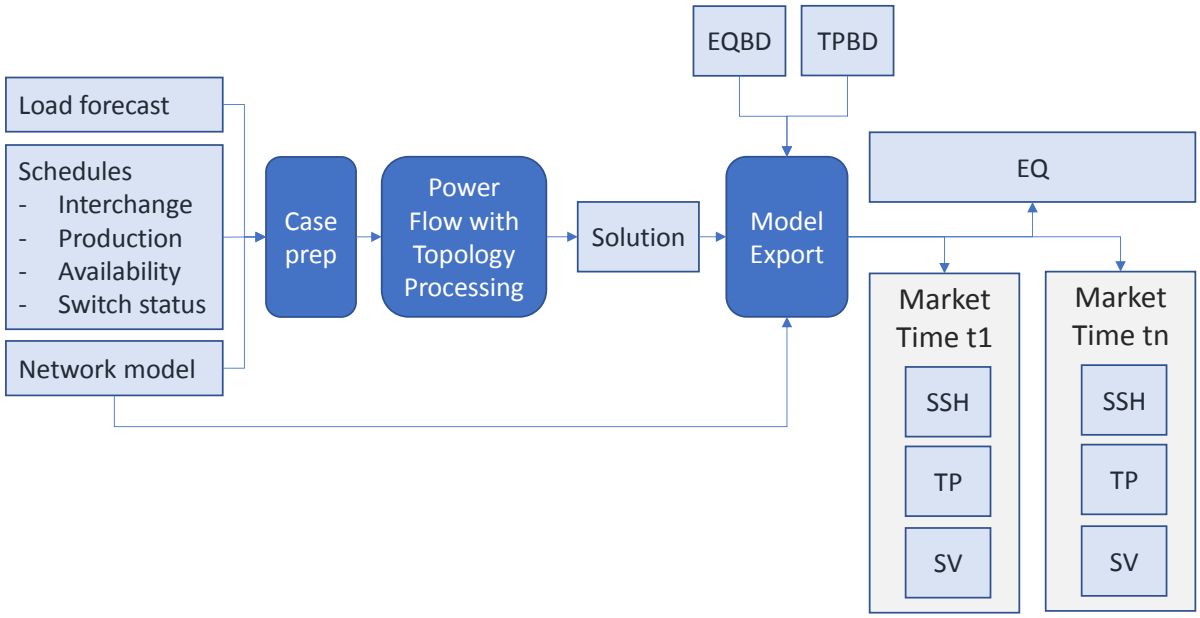



**Figure 5 Creation of IGM CIMXML files**

Inputs to IGM creation are load forecasts and market schedules shown to the left in Figure 5. From this data a power flow case is prepared for each time unit by the “Case prep” function, this corresponds to step 2 in Table 2. The case preparation also maps the schedules and load forecasts to the network model. This process is TSO’s proprietary and specific and details are outside the scope of this document.

The “Network model” to the low left in Figure 5 is created by some data management tool, typically using a proprietary information model. When an IGM is created the data from this proprietary information model need to be converted to EQ CIMXML files which is represented by the arrow from the “Network model” to the “Model export”. The “Model export” then creates all the CIMXML files as indicated in Figure 5.

Note that Figure 5 describe expected functionality and data flows between them. The functions outlined in Figure 5 can be implemented in numerous ways very different from Figure 5 so the figure should not be considered as a norm for implementations.

The power flow cases in Figure 5 are Node Breaker style and topology processing is used to compute the fictive power flow busses from the network model and switch statuses. After this, the power flow is solved and the solution including the fictive power flow busses is available as the “Solution” in Figure 5.

As discussed above, the power flow solution is usually for a power system area of interest and impact than included in the IGM. The “Model export” function in Figure 5 extracts the IGM part of the solution and adapts it to the boundary described by the EQBD and TPBD CIMXML files. Figure 5 shows the case where the same EQ CIMXML file is exported for the power flow solutions for all market time units. The solution for each market time unit is described by one SSH, one TP and one SV CIMXML file, this corresponds to step 3 in Table 2.

Once the CIMXML files has been created they are provided for validation to Quality Assurance Service provider. It may then happen that some CIMXML files fail the validation in levels 1-6 and are rejected.

If CIMXML files for a market time unit has an issue but pass the validation, e.g. due to incorrect schedule values or power flow inputs, it will made available to Merging Agent to obtain. New

CIMXML files may be created based on corrected inputs. Corrected CIMXML files shall have new md:FullModel rdf:about identifiers and md:Model.version numbers. The md:Model.version number shall be ascending and it is assumed that less than thousand revisions is needed per market time unit.

## 6.5 IGM power flow validity

An IGM power flow case is re-created by assembling its CIMXML files for a specific market time unit. The assembly uses the SV CIMXML file as starting point and trace the md:Model.DependentOn references to find the CIMXML files to include.

The CIMXML files are obtained by Merging agent, this corresponds to step 6 Table 2. According to QoCDC document, dependent SV, SSH and TP CIMXML files in an IGM shall always have the same md:Model.scenarioTime. CIMXML files for the same profile and md:Model.scenarioTime may exist with different md:Model.version numbers, if so the one with the highest version number available shall be used.

IGMs that are available have been validated as described in section 6.4.

Once IGMs have been downloaded, they are used in subsequent power flow calculations. The SSH and TP CIMXML files are used as input and initially provided original SV CIMXML is used for static validation, assembling, and can be used for cross validation of newly calculated SV CIMXML file that contains the new result as shown in Figure 6 for BB and NB models and Figure 7 only for NB models.

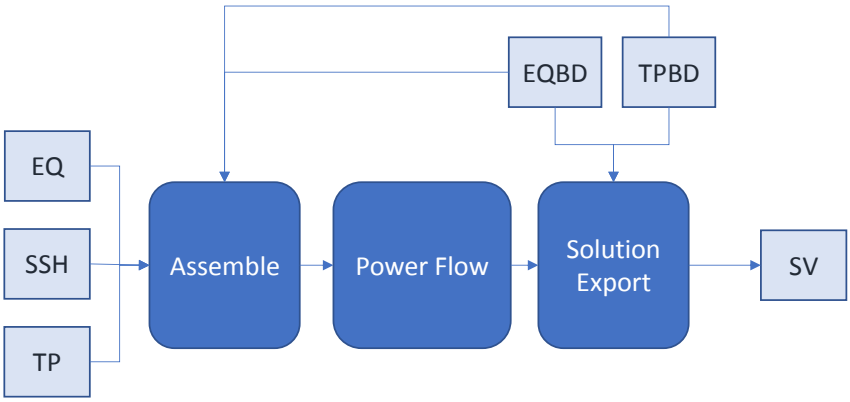



**Figure 6 Concept of solving power flow using a BB tool**

Before power flow can be solved for a BB model the EQ, SSH, TP, TPBD and EQBD CIMXML files are assembled to a power flow case. The solved power flow is exported as an SV CIMXML file.

The TP CIMXML file is a mandatory input when topology processing is not used both for NB and BB models. A discussion on BB vs. NB models can be found in section 10.5.

In general, for a power flow to be solved for a NB model, it is sufficient that the EQ, SSH and EQBD CIMXML files are assembled to a power flow case. For the current design of CGM building process, in case of NB model as described in EMF requirement specification [4] in section 3, TP CIM XML file shall also be considered as an input and the solved power flow is exported as one SV CIMXML file.

In case if potential full migration to NB models for all IGMs is accomplished, the processing of IGMs can be handled as shown in Figure 7. Note that the TP CIMXML file in the case of a NB model is an output if topology processing is used.



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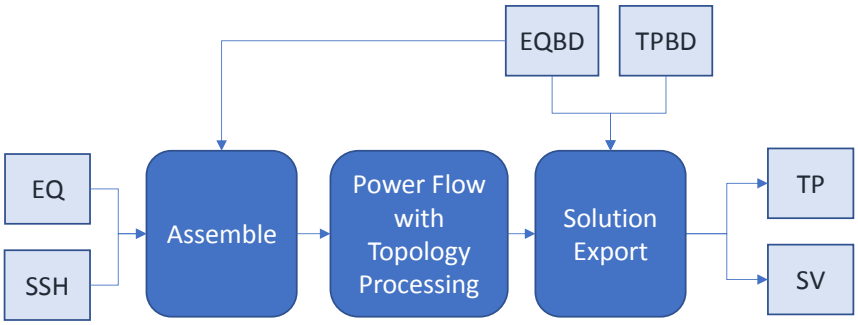



**Figure 7 Concept of solving power flow using a NB tool with topology processing capability**

The settings to use when solving power flow on an IGM both for IGM creation process and validation of a power flow plausibility by Merging Agent are described in EMF requirement specification (see [4] in section 3) section “QUALITY ASSURANCE I”, among others:

- Generation slack using cim:GeneratingUnit.normalPF as distribution factor is used to maintain area interchange (proprietary definition resulting in the same behavior in TSO’s tools as well as the “IGM Solve” path in Figure 3 for validation process).
- Voltage control is enabled, and reactive power limits shall be respected.

Given that it was possible to solve the power flow, the solution in the SV CIMXML file shall not deviate from the starting values in the SSH CIMXML file which is checked by the level 6 rules, see section 6.3.

A recommendation for a TSO to to check the quality of its IGM is to import it and rerun the power flow as shown in Figure 6 or Figure 7.

An IGM that is found valid at validation levels 1-6, shall be made available for further processing, this corresponds to steps 4 and 5 in Table 2.

## 6.6 CGM creation

A CGM is created by assembling a set of IGMs. The assembly use the SV CIMXML file for the included IGMs as starting point and trace the md:Model.DepednentOn references to find the CIMXML files to include.

The IGMs are obtained by Merging Agent, that corresponds to step 6 Table 2. The obtained IGMs are validated for plausibility by solving power flow as described in section 6.5. After this the CGM is assembled and power flow is solved for each md:Model.scenarioTime. This corresponds to step 7 in Table 2.

If IGM is not available (e.g. invalid) for a particular md:Model.scenarioTime, a search for the best available substitution is done as described in EMF requirement specification (see [4] in section 3) sections 4.6 “UPDATING OPERATING ASSUMPTIONS” and 6.6.1 “UPDATING MODEL DATA” table cell UPD9. The EMF Requirements strictly define how, in case a valid IGM for a given md:Model.scenarioTime does not exist, the IGM can be substituted after gate closure time by another IGM with different md:Model.scenarioTime.

If substitution occurs, the CIMXML files comprising an IGM used for substitution have different md:Model.scenarioTimes compared to the md:Model.scenarioTime being studied. However, the substituting IGM is treated as it is from the md:Model.scenarioTime being studied.



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As the Merging Agent is providing the updated SSH (SSHrep) (containing a md:Model.Supersedes reference to IGM's original SSH CIMXML file) and one calculated SV as a result of CGM creation process, the difference in:

- md:Model.scenarioTime of the updated SSH (SSHrep) and
- md:Model.scenarioTime of the SSH CIMXML file of the IGM used for substitution

is an indicator that the substitution of the IGM happened.

When the IGMs are created (see Figure 5), different versions of the boundary set might have been used by different TSOs. Hence, latest available official version of the boundary set is used in the CGM creation and assembly. This process requirement is due to the fact that CGM can and shall be created by using only one boundary. The validation process of IGMs with the latest official boundary set is also established to make sure that a consistent set of boundary points master data is being used and CGM can be created.

IGMs are validated against each other by checking the tie lines at the boundary TNs (see Figure 4) as follows:

- The limit values for the two cim:ACLineSegments should be consistent.
- The two cim:EquivalentInjections should either reflect transfer of power (both have non zero injection values) or reflect an interrupted tie line (both have zero injection values). A case where one cim:EquivalentInjection reflect power transfer and the other an interruption is an issue.
- The cim:BaseVoltage(s) for the two cim:ACLineSegment(s) (a retained cim:Switch or cim:PowerTransfomer is also allowed) shall not differ with more than a given limit, see QoCDC [3] for details.

Power flow for a CGM is solved without topology processing which means that TP is an input.

The inputs and outputs for power flow calculation when solving a CGM are shown in Figure 8.

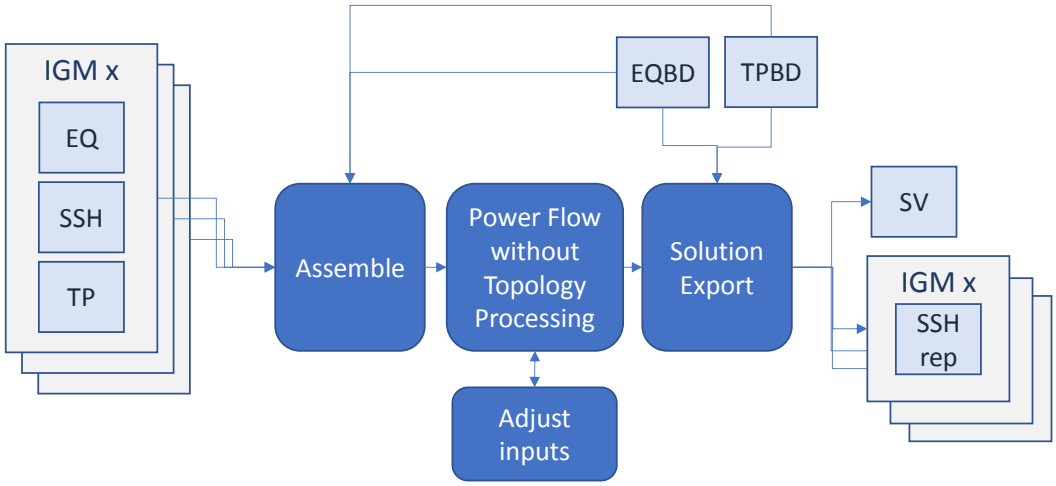



**Figure 8 Concept of solving power flow for a CGM and one market time unit**

After the CGM is assembled ("Assemble" in Figure 8), the power flow initial assumptions (SSH) might be adjusted as part of merging process and area net interchange algorithm thus new SSH CIMXML files, SSHrep CIMXML files, are created. Adjustments ("Adjust inputs" in Figure



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8) appears before the power flow is solved (e.g. aligning the status of interconnectors), while the power flow is solving (e.g. due to set limits and constraints for the power flow and area net interchange algorithms) and after the solution (aligning the SV and SSHrep values).

In case a valid IGM for the md:Model.scenarioTime time being checked was not provided before gate closure time, substitution algorithm will be applied as described previously.

The adjustments before power flow are:

- Alignment of tie lines statuses if not aligned before gate closure time, resulting in disconnecting both sides of the interconnector simulating the worst-case scenario.
- At boundary tie lines inside the CGM, see Figure 4, the injection values for both cim:EquivalentInjection(s) are set to zero so they don't impact the solution.
- For a partial assembly, that might occur and the synchronous area might have gaps in case substitution algorithm did not result in selection of an IGM for substitution (e.g. IGMs were not provided for longer period of time) boundary tie lines for the missing IGMs cannot be matched by an IGM at the remote end of the tie line. Reasons for not including some IGMs may be testing or that the IGM are remote and do not impact the solution. The flow in unmatched tie lines are represented by injection values at cim:EquivalentInjection(s) and may need adjustment as follows:
  - DC tie flows are adjusted according to market schedule if available and applicable.
  - The sum of the flows both for AC and DC at the cim:EquivalentInjection(s) (of the tie lines which are only defined on one IGM's side and for which a cim:TieFlow exists) shall match up with the sum of the net positions (cim:ControlArea.netInterchange attribute) in the assembled CGM. If this is not the case those AC cim:EquivalentInjection(s) must be scaled so they match sum of the net positions. The power factor shall be maintained when scaling. The cim:TieFlows included in the summation shall be linked to a cim:ControlArea of type interchange.

The adjustments while solving a power flow are made to respect the limits and constraints and to maintain the active power exchange for cim:ControlArea. The active power flow exchange for each cim:ControlArea shall match the net interchange as described in section 4.8 "CALCULATING THE POWER FLOW" in EMF requirement specification (see [4] in section 3). Any mismatch that appears per cim:ControlArea is distributed as load slack over the cim:ConformLoad(s) in the cim:ControlArea. The power flow is solved with settings described in section 4.9 "QUALITY ASSURANCE II" in EMF requirement specification (see [4] in section 3):

- Two different algorithms for controlling active power interchange of cim:ControlArea(s) are available:
  - Equations are embedded in the power flow solver including equations for distribution of mismatch over cim:ConformLoad(s), this is load slack.
  - A classic area net interchange that distributes mismatch between iterations over cim:ConformLoad(s) until all scheduling areas are inside the defined offset threshold per area, simulating a load slack. The residual threshold mismatch is distributed on all generating units in the CGM proportional to the reserve margin for each unit, this part of the classic algorithm is generation slack.

- Voltage control is enabled, and reactive power limits shall be respected.
- Active power control is enabled, and active power flows shall be respected.

Once the power flow is solved, the SSHrep files are exported ("SSH rep" in Figure 8) with the following adjustments and results:

- The scaled injections for cim:EquivalentInjection(s) at the borders of the CGM.
- The computed tie line power flows at the cim:EquivalentInjection(s) inside the CGM.
- The cim:ConformLoad injections resulting from load slack distribution.
- Reactive injections at cim:SynchronousMachine(s) (cim:RotatingMachine.q attribute) as a result of keeping voltage targets.
- Tap changer positions and shunt sections (cim:TapChanger.postion and cim:ShuntCompensator.section attributes) as a result of keeping voltage and active power flow targets.

The active power injections from cim:SynchronousMachine(s) are not updated in order to properly reflect the operating assumptions from the market regarding the generation schedules. The generation slack adjustments shall be small and not impact the operating assumptions.

The other values in the SSHrep files are kept the same as in the original SSH files.

The SV file include the complete solution which is the same as in SSHrep.

The power flow solutions in a CGM are created for each market time unit relevant for the business process. This means that a CGM for each studied md:Model.scenarioTime will have:

- one SV CIMXML file with a md:Model.scenarioTime representing the studied time.
- one SSHrep CIMXML file per IGM with the same md:Model.scenarioTime as the SV CIMXML file. Note that if an input IGM was substituted, SSH CIMXML file as part of the IGM used for substitution will have a md:Model.scenarioTime that differs from the studied time as recorded by md:Model.scenarioTime in the SV and SSHrep files. The replacement process is described in detail in the EMF requirements document, see [4] in section 3. The SSHrep CIMXML file in case of the substitution has a md:Model.Supersedes reference to the original SSH CIMXML file that shall be clear indication that the substitution happened.

The version number for all created SSHrep and SV CIMXML files shall be the same and start from zero.

CGM SSHrep or SV CIMXML files may need recalculation due to issues in IGM CIMXML files at one or more time units, refer to section 6.4. New CGM SSHrep and SV CIMXML files shall then be created with new md:FullModel rdf:about identifiers and md:Model.version numbers. The md:Model.version number shall be ascending and all recalculated CIMXML files for a time unit shall have the same version number.

### 6.7 Checking security of a power flow solution

Once an IGM or CGM power flow solution has been created for the base case or a contingency case it is checked against operational PATL limits. Both voltages and branch flows are checked, the rules are at validation level 8. For branches the permanent admissible transmission loading



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(PATL) limits are used. Violations indicate that the power system isn't congestion free with the given schedules and remedial actions must be taken to move the power flow solution within limits.

# 7. XML format

## 7.1 Introduction

This section gives descriptions and restrictions on the XML format.

## 7.2 The Prolog

If the standalone attribute in the prolog is missing this has the same meaning as if the standalone attribute is present with the value "yes". The meaning is that no additional document is needed to process the CIMXML file. However, the profile and quality documents are needed but that is not covered by the standalone attribute in the prolog.

## 7.3 XML name spaces

XML name spaces are used to make a vocabulary globally unique and avoid mixing it with other vocabularies. Hence an XML element or attribute name can be prefixed by an XML namespace, e.g. the word "Substation" may appear in many different vocabularies and prefixing it with a name space avoid mixing. So instead of just having the word "Substation" as element name it can be prefixed by a name space, e.g. "http://iec.ch/TC57/2013/CIM-schema-cim16#Substation". For better readability this long prefix that can be replaced by a shorter prefix "cim:Substation". When and element or attribute name with a short prefix is processed the XML software replace the short prefix with the long. To support this replacement a name space declaration shall be present in an XML file, e.g.

- xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#"

The CIM standard use of name spaces is specific to each CIM version which means that every CIM version has its own name space declaration. A consequence is that CIMXML files for different CIM versions cannot be mixed. Software reading CIMXML files need to manage the different CIM versions separately, e.g. quality gates and merging tools will have to read CIMXML files of different versions as separate sets. Hence boundary CIMXML files need to exist in one set per CIM version. Custom XML processing software might ignore this but any solution based on W3C XML standards will not.

It is technically possible to mix data described by different CIM versions by having different CIM name spaces within the same CIMXML file, e.g. a cim16 URI and cim17 URI. However, as a version of canonical CIM represents an information model consistent within that version, which means that an exchange is made for a single and specific name space.

It shall also be noted that profiles do not have their own name spaces. This makes mixing data from different profiles simple as the same name space is used given the CIM version is the same. The downside is that validation cannot use the CIM name space but must rely on the md:Model.profile element in the md:FullModel header. However, the CIMXML format (described in IEC 61970-552:2013 Ed1) is based on RDF technology and RDF compliant software natively can validate CIMXML files against profiles, or rather an ontology generated from the profile.

References to power system data described in IEC TS 61970-600-1/2:2017 and CIM have the name space prefixes:

- "cim:" for power system data, see IEC 61970-301:2016 section 4.1 [2] in section 3. Note that this name space originates from the canonical CIM and not the profiles.
- "entsoe:" for ENTSO-E extensions to power system data, see IEC TS 61970-600-1/2:2017 Ed1, [2] in section 3. Note that this name space originates from extensions of the canonical CIM and not the profiles.
- "md:" for meta data for the Model header, see IEC 61970-552:2013 Ed1, [2] in section 3.
- "dm:" for the difference model, see IEC 61970-552:2013 Ed1, [2] in section 3.
- "cgmbp:" is used for the revised meta data information model described in section **Error! R eference source not found.**.

The cim name space is updated with each release of the CIM canonical information model, e.g. cim16, cim17 etc. Hence care must be taken as the "cim:" prefix can describe different name spaces, e.g. cim16 and cim17. A reader of a CIMXML files must not mix processing of files with different name spaces and make sure to use the proper name space matching the processed CIMXML file. If not, errors will happen due to mismatch between the CIMXML file XML elements and the information model.

# 8. Metadata

## 8.1 Introduction

Restrictions on meta data and business process are described by the levels 1, 2 and 4 rules. If any of these rules fail it is not possible to create a model that can be solved by power flow. For levels 1 and 2 all information is in QoCDC [3]. For level 4 guidelines are provided in following subchapters.

## 8.2 Level 4: guidelines on dependencies

The CIMXML FullModel or DifferenceModel header describe how CIMXML file are related and the level 4 rules describe the required references between CIMXML files. The purpose is to support assembly of IGMs and CGMs. The assembly process always starts from the SV file and by tracing the dependencies starting from the SV file a complete IGM or CGM shall be obtained.

The dependencies in the CIMXML FullModel or DifferenceModel headers are checked at this level. The dependencies are described in IEC 61970-552:2013 Ed1 by the roles:

- md:Model.Supersedes, for a description see IEC 61970-552:2013 Ed1.
- md:Model.DependentOn, for a description see IEC 61970-552:2013 Ed1.

The required dependencies for md:Model.DependentOn are shown in Figure 13.

The md:Model.Supersedes is for IEC TS 61970-600-1/2:2017 restricted to the use cases:

- Update of the same limit values multiple times see example in Figure 9.
- Complete replacement of SSH files at CGM creation, see example in Figure 10.



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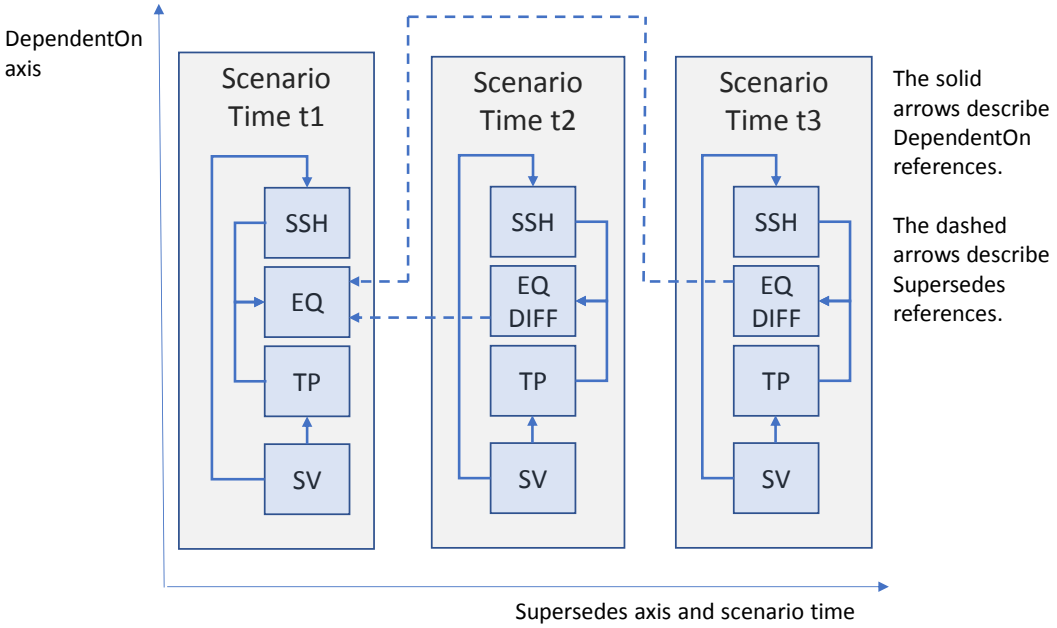



**Figure 9 Use of md:Model.DependentOn and md:Model.Supersedes in IGMs**

Figure 9 shows IGM power flow cases at three different scenario times (effectiveDateTime or md:Model.scenarioTime) and the CIMXML files for each scenario time. Scenario time T1 has CIMXML files for modelParts EQ, SSH, TP and SV and is the same as shown in Figure 13 but without the boundary files. Scenario time T2 has an EQDIFF CIMXML file instead of an EQ CIMXML file. As the EQDIFF CIMXML file at T2 describe changes to the EQ CIMXML file at T1 the md:Model.Supersedes is the reference that links them together. The complete EQ CIMXML file for T2 can be created by applying the changes in the T2 EQDIFF CIMXML file to the T1 EQ CIMXML file. The exact details will be described once a decision to implement the function is taken.

The relations between IGM and CGM CIMXML files is shown by the example in Figure 10.



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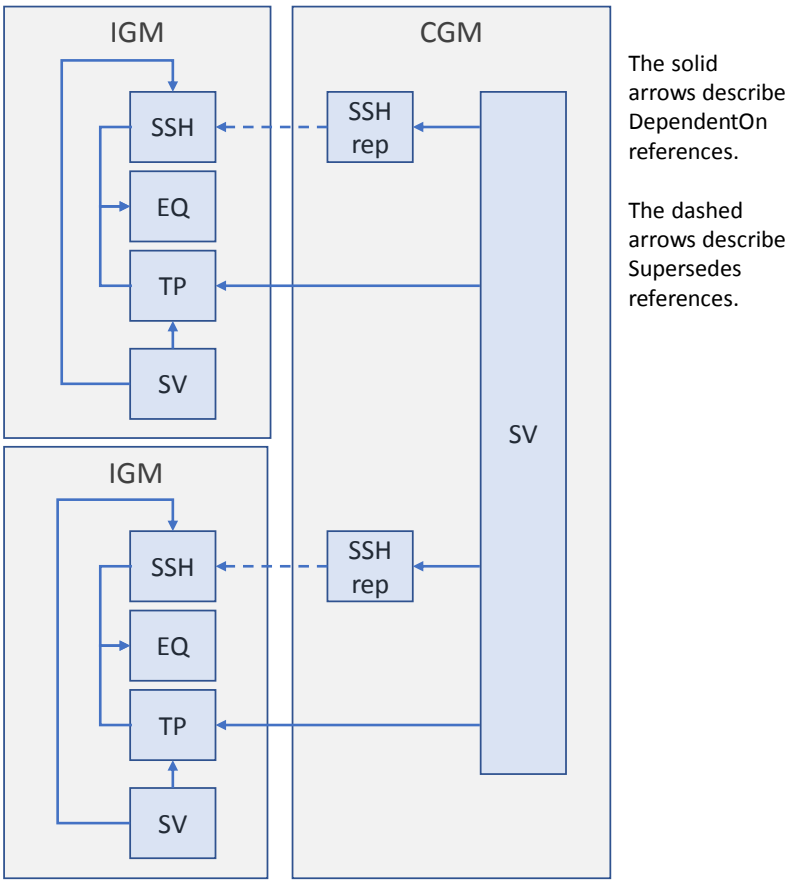



**Figure 10 Example of relations between IGM and CGM files**

Figure 10 show power flow cases for two IGMs to the left and one CGM that assemble the IGMs to the right. All three power flow cases are for the same scenario time (md:Model.scenarioTime). In case an IGM is missing for the scenario time, it can be replaced with an IGM for another scenario time as discussed in section 6.6.

For the CGM it can be noted that:

- The CGM rely on the TP and EQ CIMXML files from the IGMs.
- The CGM SV CIMXML file have one md:Model.DependentOn reference to the TP CIMXML file for each IGM.
- The EQ CIMXML files in the IGMs are located with the md:Model.DependentOn reference between TP and EQ.
- An updated SSH CIMXML file is created for each SSH CIMXML file in the IGM and they are linked with md:Model.Supersedes.

To use a CIMXML difference file, it must be applied to the CIMXML file it supersedes, i.e the difference description in the DifferenceModel element is applied to the superseded CIMXML file and the operations to apply in general are:

- Addition of new objects
- Deletion of existing objects

- Update of attribute values

Note that this specification only supports using the DifferenceModel for limit value updates (changes) in EQ.

These operations result in a new CIMXML file that contains the combination of superseding and the superseded files as shown in Figure 11.

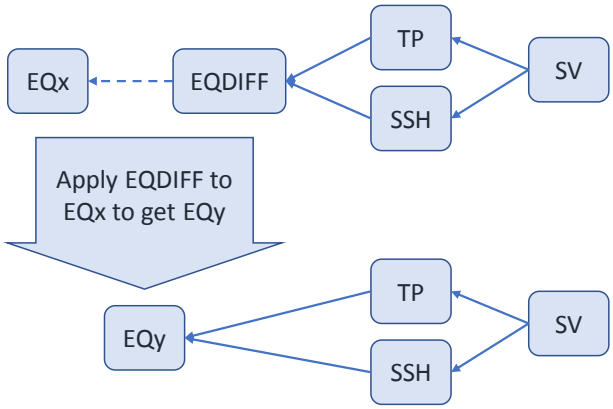



**Figure 11 Application of DIFF files**

In Figure 11 the EQ CIMXML file EQx is Superseded by the EQDIFF CIMXML file. Applying the differences results in a new EQ CIMXML file EQy. EQy has the same meta data as EQDIFF which means the mRID, scenario time, description, profiles, MAS etc. are the same. The mRIDs being the same means that the rdf:about attribute in the headers are the same. Hence the DependentOn references from TP and SSH to the original EQDIFF are not affected and work with the new EQy.

The discussion above indicates that the application of the difference EQDIFF is made with the files, but this is just one option. Places where a difference CIMXML file can be applied are:

- Before uploading CIMXML files to a middleware which means that no difference files are uploaded. This option is not used.
- By the middleware after upload as shown in Figure 12.
- By the receiving client after download from the middleware. This option is not used.

Figure 12 show the application of an EQDIFF CIMXML file to the EQx file by the middleware (OPDE in this case) which means that Receivers don't see the EQDIFF CIMXML file but only the resulting EQy CIMXML files.



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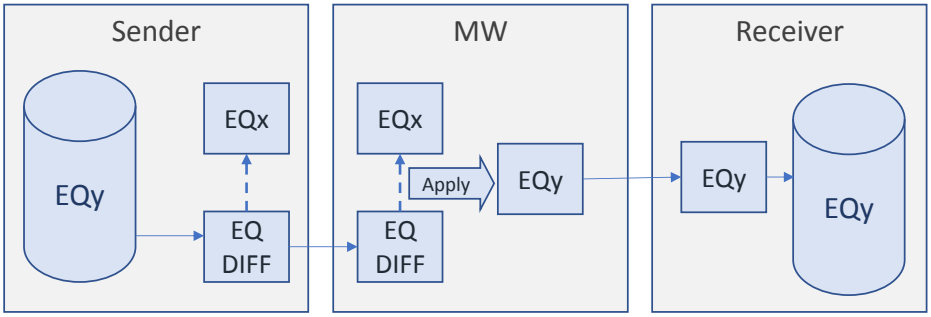



**Figure 12 Applying the EQDIFF at the Middleware**

In Figure 12 the EQDIFF CIMXML file is transferred to the middleware (OPDE) where it is applied to the EQx CIMXML file referenced by the md:Model.Supersedes in the difference file. This result in the EQy CIMXML file that is downloaded by the Receivers.

Applying difference files at the middleware is also required for the validation as EQDIFF CIMXML files cannot be validated on their own.

The completeness of data across CIMXML files are also checked at level 4 which include two types of checks:

- The references between objects resolve, i.e. an object referenced by a rdf:resource exists.
- Objects are complete meaning that all mandatory attributes or references are present according to cardinality rules in the information model.

Figure 13 show the required dependencies (also in QoCDC [3]). Note that more references than the ones shown in Figure 13 exists between objects in different CIMXML files. The references shown in Figure 13 are the ones needed to successfully assemble and IGM or CGM.



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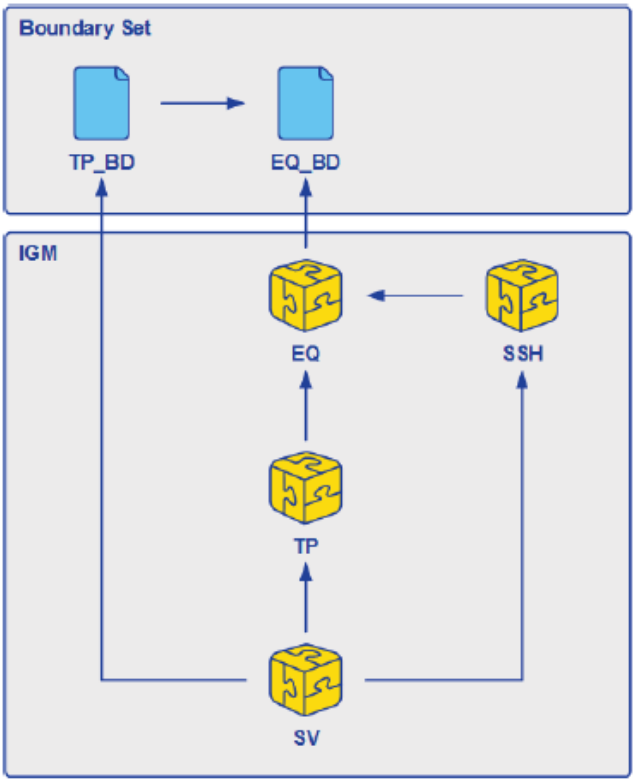



**Figure 13 CIMXML file md:Model.DependentOn dependencies**

When a model is being assembled, it starts with locating the SV file and collecting the files referenced directly or indirectly from it.

Note that the boundary references in the SV and EQ files relate to the boundary files used when the IGM or CGM was created. When creating a CGM the latest available boundary files shall be used. It could then be that these boundary files are different from the one used in the IGM.

# 9. Boundary data

## 9.1 Introduction

To create an IGM as described in section 6.4 or CGMs as described in section 6.6, it is necessary to have a boundary that describes the connection with other IGMs. The boundary is described in the CIMXML files EQBD and TPBD.

A simplified view how a cim:TopologicalNode (TN) in the boundary CIMXML file TPBD connects IGMs together is shown in Figure 14. For a boundary to work properly this simplified view is extended with information describing power net interchanges between the IGMs or rather control areas within IGMs assembled into the CGMs.

This section further elaborates the specifics of the boundary.

## 9.2 Incomplete connectivity at boundary

An IGM is incomplete and a power flow cannot be solved with an IGM alone. By assembling an IGM with the boundary, the IGM becomes complete and power flow can be solved. This boundary situation is shown in Figure 14.



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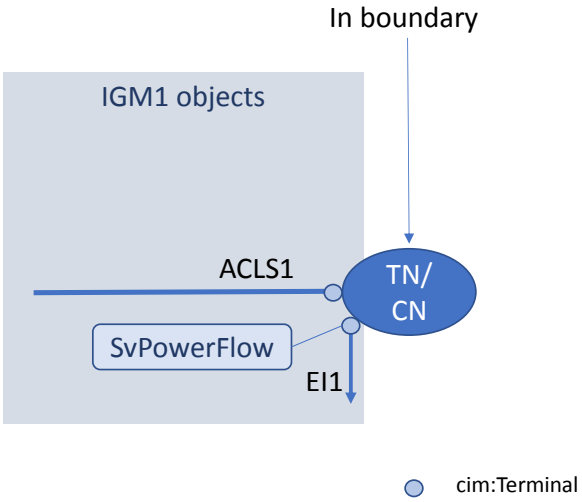



**Figure 14 Boundary with a single IGM**

The IGM connects to a TN and CN with a cim:Terminal belonging to one branch, typically a cim:ACLineSegment, and one cim:EquivalentInjection. The cim:EquivalentInjection's cim:Terminal has a cim:SvPowerFlow.

The cim:SvPowerFlow injection represents the power flow in or out of the neighbouring IGM at the other side of the boundary. This is how the power flow for an IGM can be solved according to section 6.5.

## 9.3 Complete connectivity at boundary

A CGM having two IGMs connecting at a boundary TN/CN is shown in Figure 15.

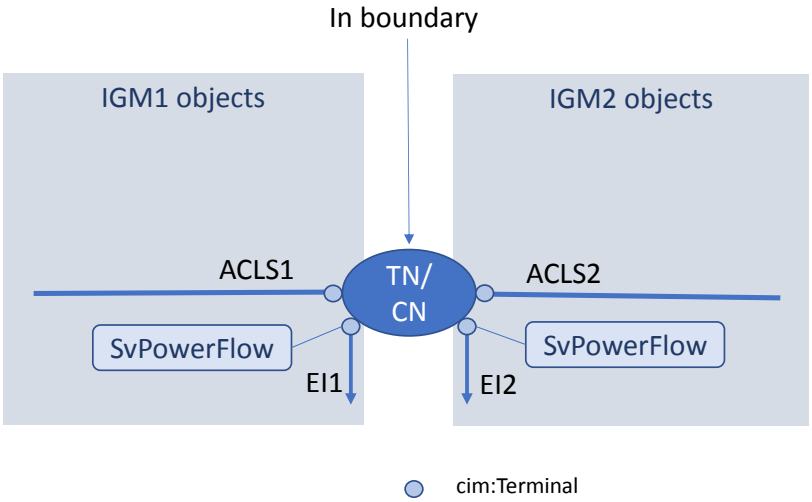



**Figure 15 Boundary with two IGMs**

Each IGM connects to the TN/CN with cim:Terminal(s) belonging to one branch, a cim:ACLineSegment in this case, and one cim:EquivalentInjection. The cim:EquivalentInjection's cim:Terminal has a cim:SvPowerFlow.

The sum of the power flows in EI1 and EI2 are supposed to be equal with different signs but as this typically is not the case, they shall be set to zero as part of the CGM creation process described in section 6.6.



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If the power flow is zero for one cim:EquivalentInjection but non-zero for the other this indicates that the TSOs has different opinions on the tie line operational status which show poor coordination between the TSOs. This situation may occur if the side with zero injection is disconnected, e.g. by a cim:Switch.status or cim:ACDCTerminal.connected flags.

## 9.4 Control area conventions

When a tie line, e.g. a cim:ACLineSegment, is connected to a boundary TN/CN which is also a boundary for a cim:ControlArea of type interchange (cim:ControlArea.type = cim:ControlAreaTypeKind.Interchange) it may have a cim:TieFlow. If a tie line is included in transfer capacity trade it shall have a cim:TieFlows as shown in Figure 16.

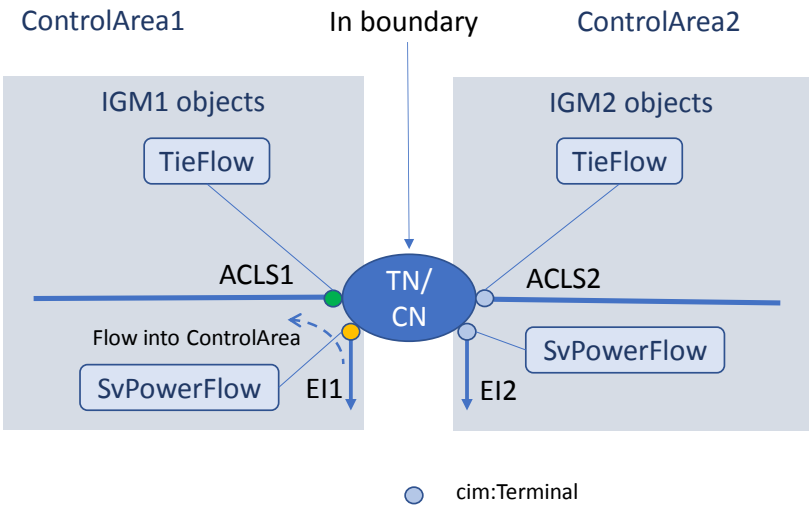



**Figure 16 Transfer capacity trade at boundary with two IGMs**

The flows at cim:Terminal(s) (e.g. cim:SvPowerFlow) follow the load sign convention, i.e. flow from the TN into the cim:Terminal is positive. This means that a power flow at a tie line connected to a boundary TN is a flow into the cim:ControlArea. Figure 16 shows an example where the dashed arrow represents a flow into the cim:ControlArea and the ACLS1 cim:Terminal, shown with green in Figure 16, has a positive flow which is the same sign convention as for cim:ControlArea.netinterchange.

The calculated power flow for the tie line is updated at the cim:SvPowerFlow at the cim:EquivalentInjection as tie line is not required to have a cim:SvPowerFlow. Figure 16 shows a cim:SvPowerFlow linked to the EI1 cim:Terminal, yellow in Figure 16. The flow at the EI1 cim:Terminal is into the TN which is the negated value of the tie line flow.

The flag cim:TieFlow.positiveFlowIn is true if the flow from the TN into the tie line cim:Terminal, green in Figure 16, is also a flow into the cim:ControlArea. But with the conventions in Figure 16 where the cim:TieFlow is located at the tie line end towards the boundary and the cim:SvPowerFlow is at cim:EquivalentInjection the cim:TieFlow.positiveFlowIn being located at the tie line cim:Terminal has lost its meaning.

For trade with an IGM not included in the CGM a cim:TieFlow is still present at the boundary TN but no tie line will be connected for the not included IGM, see Figure 17.

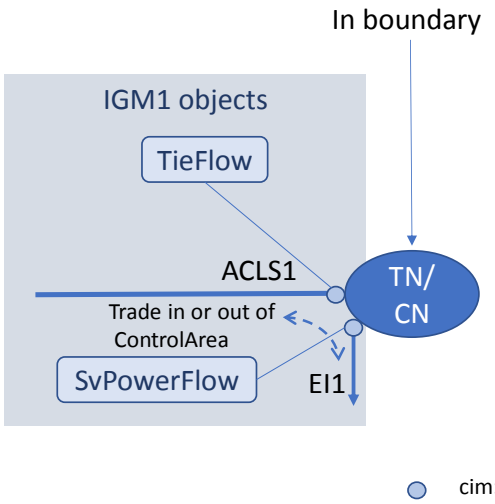


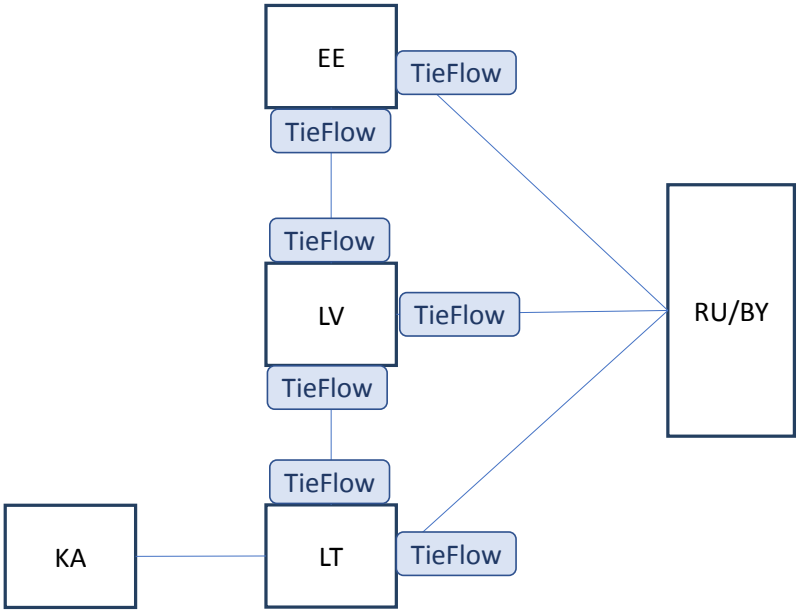

**Figure 17 Trade with at a boundary with single IGM**

The cim:EquivalentInjection.p value, see dashed double arrow in Figure 17, describes the flow to or from the not included IGM and must have a realistic value which is the reasoning in case of partial merge for scaling the boundary injections (of the tie lines which are only defined on one side and for which a cim:TieFlow exists) to match the net interchanges as described in section 6.6.

A summary of the control area conventions defined by CGM Building Process Working Group for the purpose unification of that shall enable sustainable CGM Building Process is the following:

- cim:TieFlows shall be present at the tie line ends connected to a boundary TN in case the tie line carry traded energy and included in the net position (cim:ControlArea.netInterchange). The sum of the tie flows equals the net position of the control area (cim:ControlArea.netInterchange) that is be compared towards the sum of AC and DC trades.
  - In the case of direct tie lines between a TSO and its neighbouring TSOs for which the trade is included in the AC net position cim:TieFlow(s) shall be present as shown in Figure 16. Lower level voltage tie lines (132kV, 150kV, …) can have a cim:TieFlow depending on how the TSO assess the realisation of the sold capacity in real-time operation.
  - If the sold capacity is assessed and measured in real-time over the 400kV and 220kV lines but not over the lower voltage tie lines (e.g. 132kv and 150kV), then those lower voltage level lines shall not have any cim:TieFlows associated.
  - If sold capacity is assessed and measured also over lower voltage tie lines (e.g. 132kV and 150kV) in the real-time, then cim:TieFlow shall be defined for those lines.
  - In case of a partial merge some boundaries will have the situation shown in Figure 17. A variation of this case is if two TSOs (say A and B) having energy trade where traded energy flow is transferred through yet another TSO (say X) not included in the trade. An example is trade between Elering and AST (being TSO A and B) where parts of the traded energy is transferred between them via Russia (being TSO X), see Figure 18. Then
  - The trade shall be included in the AC net positions for the trading TSOs (A and B).

- The boundary between the trading TSOs and the transfer TSO shall have a cim:TieFlow at the trading TSO side as shown in Figure 17 and Figure 18.
- TSO must use pre-aligned and validated cim:EquivalentInjection values for RSCs to obtain a correct assembled model after applying interchange control.
- Merging tools shall assess the sum of the flows both for AC and DC at the cim:EquivalentInjection(s) (of the tie lines which are defined on one side and for which a cim:TieFlow exists). This shall match up with the sum of the net positions (cim:ControlArea.netInterchange attribute) in the assembled CGM. If this is not the case those AC cim:EquivalentInjection(s) must be scaled so they match sum of the net positions. The power factor shall be maintained when scaling. The logic is like the case of partial merging.




The TieFlow rectangles show cim:TieFlow instances at the tie lines.

**Figure 18 Example of trade via a TSO RU/BY**

- In case two TSO include a tie line (they both refer to the same boundary TN), both TSO shall include cim:TieFlows. If cim:TieFlows are not included none of the TSOs shall include cim:TieFlows. The case where one TSO include a cim:TieFlow but not the other is erroneous.
- No cim:SvPowerFlow is required at tie line ends.
- cim:SvPowerFlows shall be present at the cim:EquivalentInjection connected to the same boundary TN as the tie line and have the negated tie flow.
- The attribute cim:TieFlow.positiveFlowIn shall always be true.
- For a cim:ControlArea the sum of cim:ControlArea.netinterchange and the cim:EquivalentInjection cim:SvPowerFlow.p for tie flows, branches having a cim:TieFlow, shall sum to zero. If this is not the case power is leaking due to inconsistently defined cim:TieFlow(s).



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- HVDC tie line included in energy trade shall be included in a cim:ControlArea, have a cim:TieFlow and be included in the cim:ControlArea.netinterchange.

- In case the of an partial assembly (refer to section 6.6) there is a need to scale cim:EquivalentInjection(s) of tie lines which are defined on one side and for which a cim:TieFlow exists, the mismatch should be spread proportionally over those cim:EquivalentInjection(s): commercial exchanges on borders cannot be used because of loop flows.

- A TSO may define multiple cim:ControlArea but only one of type Interchange. Only cim:TieFlows associated to the cim:ControlArea of type Interchange shall be considered to assess the net position of a cim:ControlArea. This is in line with rule NetInterchange2 of QoCDC [3]. In Figure 19, 3 cim:TieFlow are considered in the net position:

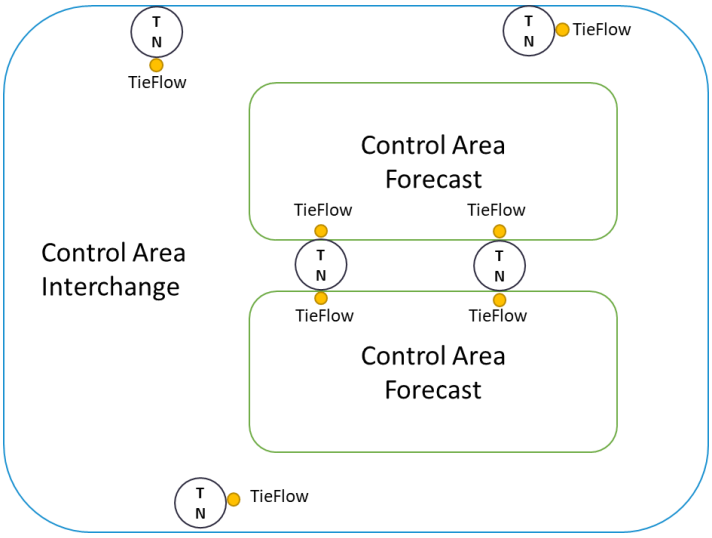



**Figure 19 cim:TieFlow example**

## 9.5 Boundary containment

The boundary TN and CN is contained by a cim:Line in the boundary, see Figure 20.



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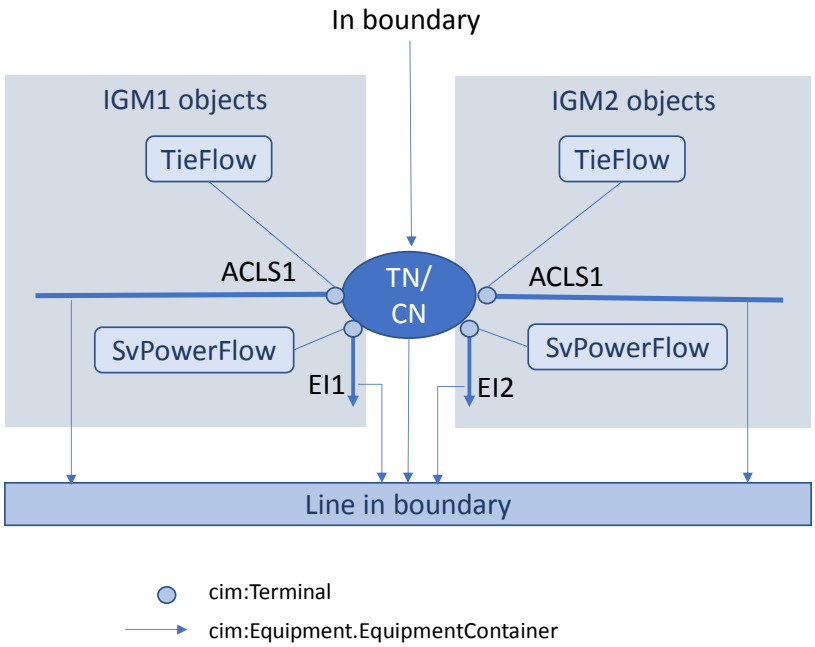



**Figure 20 Boundary containment**

The IGM cim:ACLineSegment and cim:EquivalentInjection should also be contained by the cim:Line in the boundary to avoid creation of unnecessary container objects in the IGMs.

## 9.6 Boundary Configuration Examples

The connections to a boundary CN or TN described in previous sections use two cim:ACLineSegment(s). As shown in Figure 20, the cim:ACLineSegment(s) are contained by a cim:Line. A cim:ACLineSegment can only be in a cim:Line, see rule LineContainment, but according to rule SvPowerFlowBranchInstances, a cim:Switch or cim:PowerTransformer may also connect to a boundary CN/TN. This section shows a couple of allowed boundary configuration cases.

The figures below show cim:Substation(s), Substation1 to the left in ModelingAuthoritySet (MAS) MAS1 and Substation2 to the right MAS MAS2. In the middle there is a cim:Line Line1-2 coloured in red and from the ENTSO-E boundary. The cim:Equipment and CNs/TNs drawn inside the containers Substation1, Substation2 or Line1-2 are contained by that container. The cim:Equipment and CNs/TNs belongs to the MAS or boundary shown in the background. Line1-2 contains a cim:ConnectivityNode CN1-2.

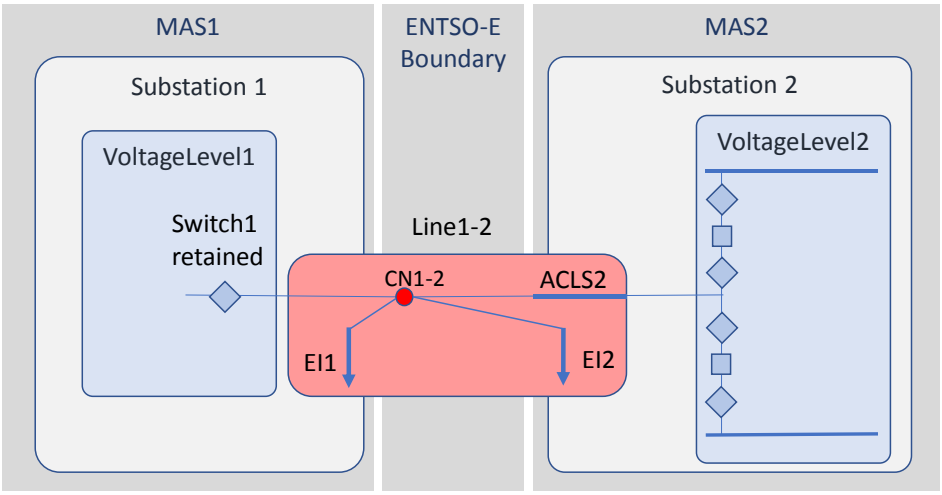



**Figure 21 Boundary with cim:Switch**

Figure 21 show how a cim:Switch can connect to the boundary instead of a cim:ACLineSegment. A cim:Switch is not allowed in a cim:Line but it can be within a cim:VoltageLevel. Note that the cim:Switch must be retained. The containment and MAS belongings are as follows:

- Line1-2 belongs to the ENTSO-E boundary.
- CN1-2 is contained by Line1-2 and belongs to the ENTSO-E boundary.
- EI1 is contained by Line1-2 and belongs to MAS1.
- Switch1 is contained by VoltageLevel1 and belongs to MAS1.
- EI2 and ACLS2 are contained by Line1-2 and belongs to MAS2.

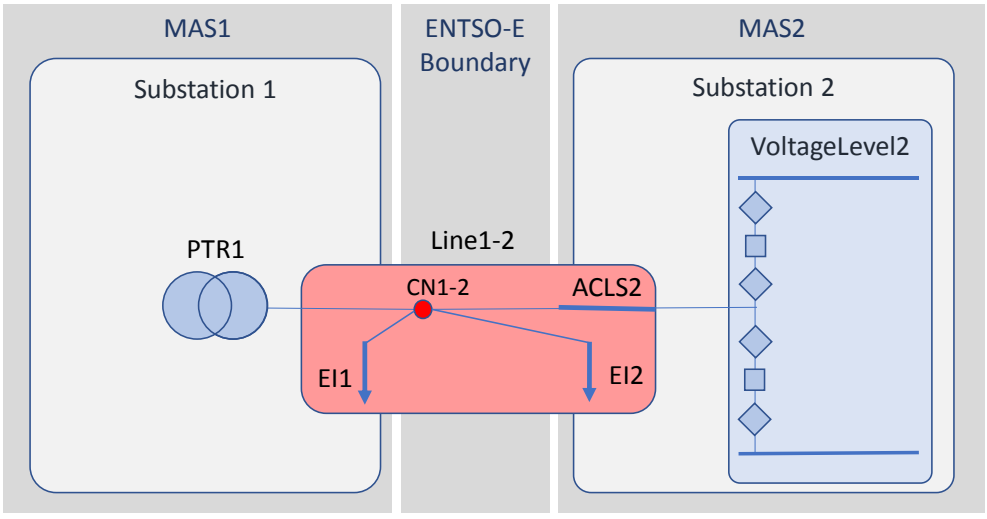



**Figure 22 Boundary with cim:PowerTransformer**

Figure 22 show how a cim:PowerTransformer can connect to the boundary instead of a cim:ACLineSegment. A cim:PowerTransformer is not allowed in a cim:Line but it can be within a cim:Substation. The containment and MAS belongings are the same as in Figure 21 except

for:

- PTR1 that is contained by Substation1 and belongs to MAS1.

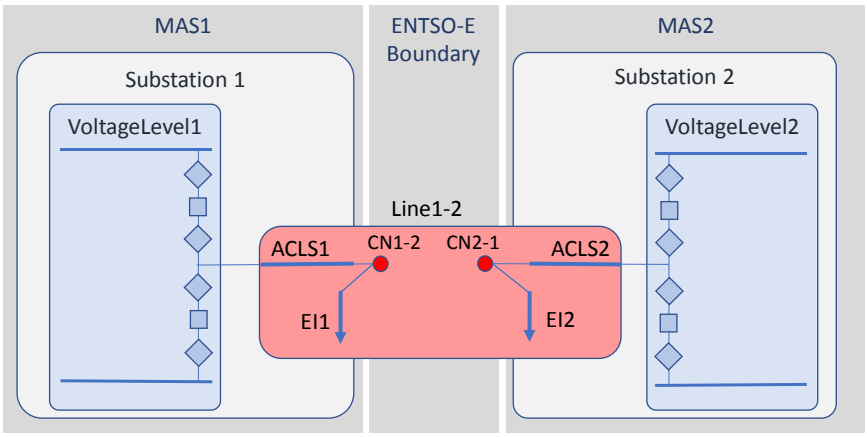



**Figure 23 HVDC boundary with cim:ACLineSegments**

Figure 23 show a HVDC boundary with two cim:ACLineSegments. Note that the Line1-2 in this case has two CNs, one at each side of it. A difference between the injection values at EI1 and EI2 represents the losses in the HVDC line. The containment and MAS belongings are as follows:

- Line1-2 belongs to the ENTSO-E boundary.
- CN1-2 and CN2-1 are contained by Line1-2 and belongs to the ENTSO-E boundary.
- EI1 and ACLS1 are contained by Line1-2 and belongs to MAS1.
- EI2 and ACLS2 are contained by Line1-2 and belongs to MAS2.

Note that ACLS1 and ACLS2 are fictitious and do not exist in a real HVDC line. Hence a better model is shown in Figure 24.

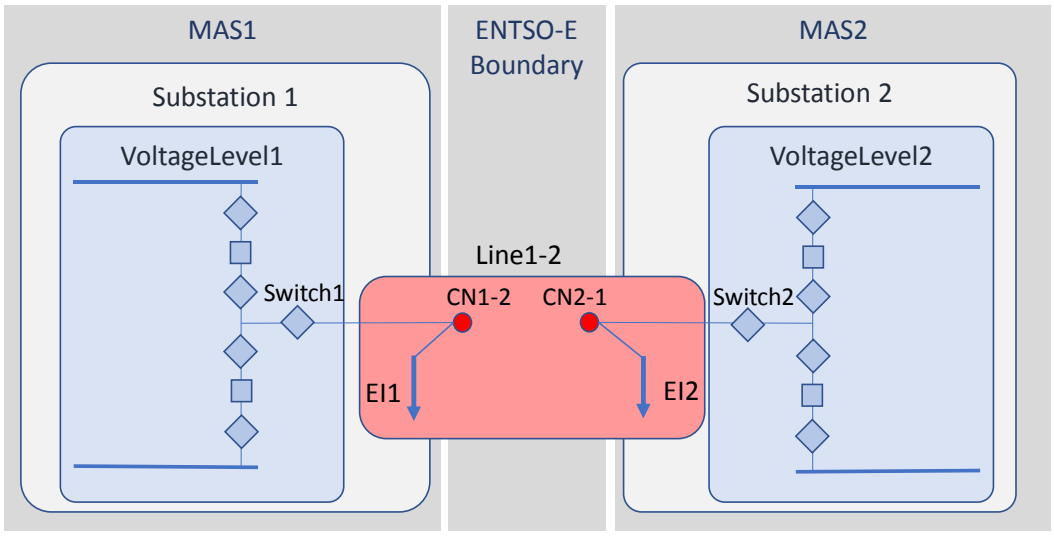



**Figure 24 HVDC boundary with cim:Switches**

Figure 24 show a HVDC boundary with two cim:Switches. It is the same as for Figure 23 with the following differences:

- EI1 is contained by Line1-2 and belongs to MAS1.
- Switch1 is contained by VoltageLevel1 and belongs to MAS1.
- EI2 is contained by Line1-2 and belongs to MAS2.
- Switch2 is contained by VoltageLevel2 and belongs to MAS2.

Switch1 and Switch2 shall both be retained and may or may not be fictitious depending on how the substation is physically built. Note that each of the two CNs have a corresponding TN in the boundary. A topology processor shall make sure to use this TN. The reason Switch1 and Switch2 are retained is to make this possible.

### 9.7 BaseVoltage

The attribute cim:BaseVoltage.nominalVoltage describes a system wide base voltage. Hence the number of cim:BaseVoltage(s) is way less than all the possible rated voltages in the system and it is preferable not to create a base voltage for every rated voltage, e.g. a SynchronousMachine with a rating cim:RotatingMachine.ratedU = 16.1 kV still operates at the system voltage 16 kV. For this reason, the cim:BaseVoltage(s) shall be agreed and be present in the boundary to avoid excessive use of cim:BaseVoltage(s) having small differences.

The only object types that can connect cim:BaseVoltage(s) with more than 10% voltage difference are cim:EquivalentBranch or cim:PowerTransformer.

## 10. Network data

### 10.1 UML model issues

Information modelling in UML can be done in many ways. One way is to be strict on cardinalities which tend to create a more complex UML or use open cardinalities, e.g. [0..*], that allows for a simpler UML. Downside with the latter is that strictness needs to be added later as done with the rules in QoCDC ([3] in section 3). The complexity avoided with the simplified UML then comes back in the form of way more complex rules at worst described in English text or better described in a formal language as Object Constraint Language (OCL).

The reason for the many rules needed is oversimplified UML modelling.

For the canonical CIM, IEC 61970-301:2016 ([1] in section 3), the modellers made the choice of a single and deep inheritance hierarchy. For equipment see example in Figure 25.



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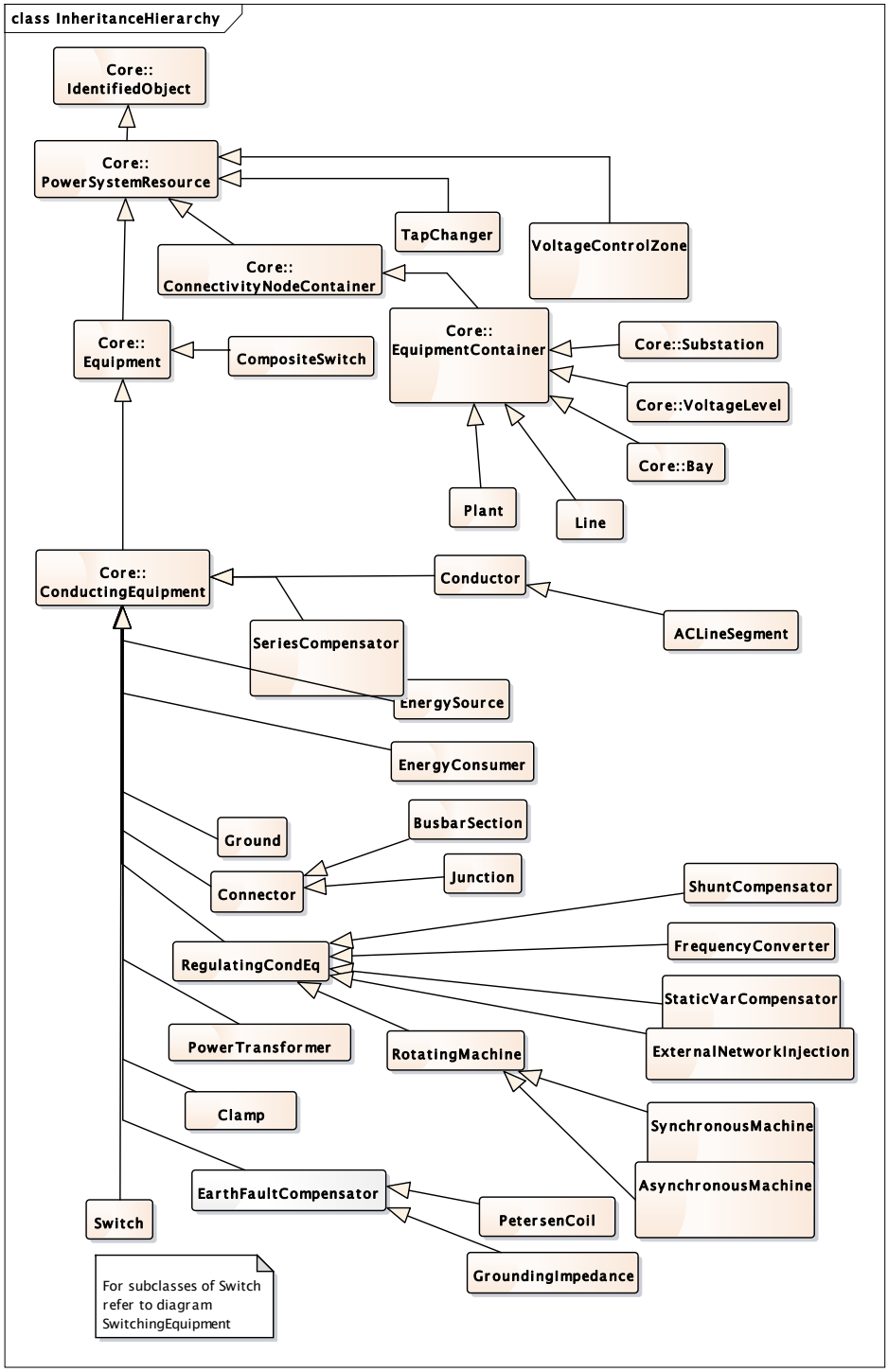



**Figure 25 Canonical CIM equipment inheritance hierarchy**

A deep inheritance structure as in Figure 25 has the benefit of non or only little redundancy, i.e. data is only described once. An example is the cim:Equipment-cim:EquipmentContainer association shown in Figure 26.

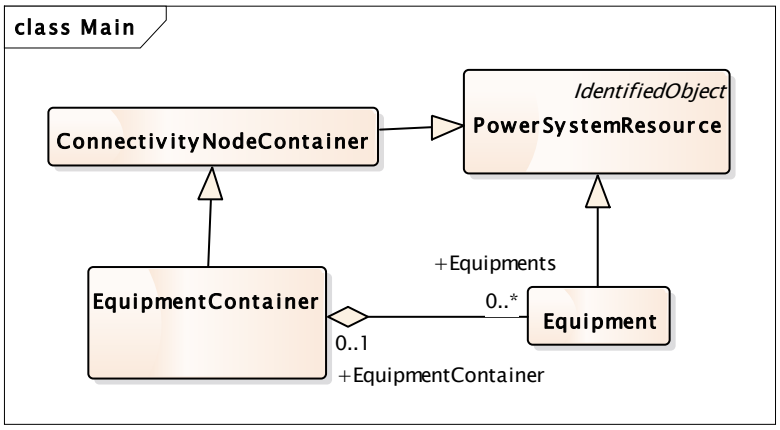



**Figure 26 The Equipment-EquipmentContainer association**

The association in Figure 26 is inherited by all subtypes of cim:Equipment (see Figure 25) and cim:EquipmentContainer (cim:Substation, cim:Line, cim:VoltageLevel etc.). This allows to put any of the cim:Equipment subtypes in any of the cim:EquipmentContainer subtypes where many combinations don't make sense, e.g. cim:SynchronousMachine in cim:Line, cim:ACLineSegment in cim:Substation etc. There are numerous examples of this in canonical CIM, another is the cim:Terminal-cim:ConductingEquipment association.

Another common issue with the canonical CIM is when combinations of attribute values do not match with the class:

- Some cim:RegulatingControl.mode attribute values do not make sense for some of the subclasses of cim:RegulatingControl, e.g. the activePower control mode is only relevant for phase tap changer transformers.
- For the ENTSO-E extension cim:OperationalLimitType.limitType only a few of the combinations of the limitType values and cim:OperatinalLimit subtypes make sense, e.g. the PATL type is only relevant for branch flows.
- A cim:Switch has two cim:Terminals. The attributes cim:Switch.open and cim:ACDCTerminal.connected describe the connectivity status which means there are three ways to interrupt power flow through a switch by using of the open flag or one of the two connected flags.

There are many more examples of this issue.

Yet another issue is when a class is overused to describe unrelated usages, e.g. use of cim:EquivalentInjection (EI) to represent a reduced injection, a boundary injection or an HVDC link. The three cases need to be distinguished from each other and this is done as follows:

- Reduced injection; no specific indication and original use case.
- The flag TN.boundaryPoint where the EI is connected tells the EI represents a boundary injection.
- The text "HVDC" at the TN where the EI is connected tells the EI represents a HVDC link.

A better information model would have been to use a dedicated class for each case.

When this happens, it is an indication that new and more specific classes are need.

It is possible to create UML models without the above issues but with the consequence of more complex UML models.

A way to restrict a too open UML model is to use rules described in the Object Constraint Language (OCL) that works together with the UML information model. This increases the complexity of the information model and if CIM was to be refactored in the future the trade-off between a more restrictive class model in UML or use of OCL rules could be investigated.

The least intrusive way to restrict the current CIM information model is to use OCL rules which is extensively used for the IEC TS 61970-600-1/2:2017 profile ([1] in section 3). The restrictions are described in the QoCDC document ([3] in section 3).

## 10.2 Rules

### 10.2.1 Overview

Rules that describe the quality of network data can be divided in groups describing increasingly complex relations in data:

A. Most basic are schema rules defined by the profiles derived from canonical CIM UML. The profiles are formally expressed in an XML format described in IEC 61970-501:2006 [1] in section 3.

B. Restrictions of values that cannot be described in UML, e.g. relations as x<y and string lengths.

C. Additional restrictions on too open role cardinalities, e.g. switches shall have exactly two cim:Terminal(s).

D. Additional restrictions on combinations of data where role references are followed to collect and cross check data from different classes. Several of these restrictions are defined in IEC TS 61970-600-1/2:2017 ([2] in section 3) but many new restrictions have been added based on experiences from interoperability tests.

IEC 61970-501:2006 is based on RDFS that is an extended and non-standard XML format based on the RDFS specification:

- www.w3.org/TR/1999/PR-rdf-schema-19990303

The specification is outdated and since 1999 both the OWL specification and the RDFS 1.1 specification from 2014 has been created. The custom IEC 61970-501:2006 extensions are now covered by the latest OWL and RDFS specifications and IEC 61970-501:2006 needs to be revised.

Validation of network data is divided in levels as described in section 6.7. The above groupings are not explicitly referred to by the rules defined in QoCDC [3] in section 3.

Validation is done in the two steps “CIMXML file validation” and “Model validation” shown in Figure 3.

“CIMXML file validation” is done within the scope of the CIMXML file and includes:

- Level 2 profile schema validation according to group A above. This means checking that class, attribute and role names in XML elements match the profile schema and that cardinalities are respected.
- Level 3 additional validation according to groups B, C and D. Groups C and D use local associations within a CIMXML file.

“Model validation” is done for an assembled model and includes level 5 and up in Figure 3.



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The rules have a scope according to groups C and D above. Groups C and D use data not possible to check within a single CIMXML document.

### 10.2.2 Levels 2 and 4, profile schema related rules

These rules report issues related to the network profile schema, see Table 5. These rules appear at level 2 and 4.

**Table 5 Profile schema rules**

| Rule | Level | Schema deviation |
|---|---|---|
| NotMandatoryClass | 2 | Class name deviations. |
| NotMandatoryProperty | 2 | Attribute and role name deviations. |
| DanglingReference | 4 | A reference cannot be resolved in assembled model. |
| IncorrectAttributeOrRoleCard | 4 | Cardinality violated for attributes or roles |

For the rule IncorrectAttributeOrRoleCard note that the OCL language only check cardinalities for roles and string values. Missing numeric values are defaulted to zero so they are always present as zero values regardless of cardinality. This is a flaw in the OCL language.

### 10.2.3 Level 3 rules

Tables showing rule references to the information model are shown in Annex 2 – References to attributes and roles from QoCDC rules.

### 10.2.4 Level 5 rules

There are three rules restricting how cim:SvPowerFlow is used:

- SvPowerFlowBranchInstances with boundary related restrictions, an ERROR
- SvPowerFlowInstances with class related restrictions, an ERROR
- SvPowerFlowBranchInstances2 with class related restrictions, a WARNING

The purpose of the two first rules is to ensure the power flow result is made available, hence they are required.

The purpose of the last rule is to make the power flow result for branches available so the solution can be reviewed at any branch. This is highly wanted when solutions for the same IGM or CGM but from different tools are compared. The lack of this information has resulted in development of scripts calculating the flows which requires time which is not applicable in the operational process if the comparison is necessary. The amount of information that the additional cim:SvPowerFlow instances will add in the SV CIMXML files is approximately two times the number of branches. A rough estimate is 50% growth, not a disaster and a way less the size of EQ CIMXML files.

Tables showing rule references to the information model are shown in Annex 2 – References to attributes and roles from QoCDC rules.

### 10.2.5 Level 6 rules

Tables showing rule references to the information model are shown in Annex 2 – References



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to attributes and roles from QoCDC rules.

#### 10.2.6 Level 7 rules

Tables showing rule references to the information model are shown in Annex 2 – References to attributes and roles from QoCDC rules.

#### 10.2.7 Level 8 rules

Tables showing rule references to the information model are shown in Annex 2 – References to attributes and roles from QoCDC rules.

### 10.3 Containment

The physical equipment in a power network is organised in a structure starting with geographical regions having substations having bus bar systems having equipment and so on. IEC TS 61970-600-1/2:2017 enforces a containment structure suiting transmission network. The reasons for IEC TS 61970-600-1/2:2017 doing this are:

- It keeps fidelity with how transmission networks are built.
- It enables recognition of objects across different types of models. In a bus-branch model that do not include substations, the bus names may imply the substation, which makes mapping between simplified and detailed models difficult.
- Tools commonly rely on the containment structure for model navigation.

The substation is an important concept that describe equipment located in fenced area of limited size. Substation are electrically connected with other substations by transmission lines.

The bus bar systems within a substation is called cim:VoltageLevel and include the bus bars (cim:BusbarSection in CIM) and equipment.

In a real substation all equipment is included in bays but in CIM only switches are in a cim:Bay.

Some misuses of substation containment are:

- A single cim:Substation for the whole network.
- One cim:Substation per busbar system which means that every cim:Substation include exactly one cim:VoltageLevel despite transformers in the cim:Substation resulting transformers spanning multiple substations.

### 10.4 Disconnecting switches

Switches can be disconnected with the following three attributes in series:

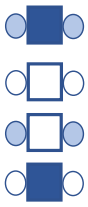
- The cim:Switch.open flag
- Either or both cim:ACDCTerminal.connected flags

This allows for variation, as shown in Figure 27, that complicates solutions.



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1. Closed path
2. Open path Switch open and ACDCTerminal connected
3. Open path Switch open
4. Open path ACDCTerminal connected



- Options for less variations on Switches

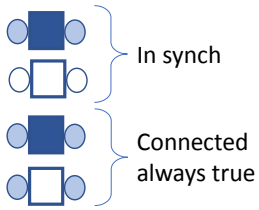



**Figure 27 Combination of cim:Switch.open and ACDCTerminal.connected flags**

Hence the cim:ACDCTerminal.connected flag for cim:Switch(es) shall always be true. This avoid caring about these flags for cim:Switch and its subclasses.

Level 5 rule SwitchOpenVsConnected also checks this.

## 10.5 Node-breaker vs. bus-branch modelling

Modelling a power network can be made at two levels of detail:

- Node-breaker
- Bus-branch

Bus-branch models are simpler than node-breaker models which is beneficial in planning scenarios while node-breaker models are used in real time systems (SCADA/EMS) where the power flow input and result is mapped to real equipment. Once a detailed node-breaker model has been created there is no reason to have a bus-branch model maintained in parallel with the node-breaker model. This result in extra work keeping the duplicate models in synch. Better is to derive the bus-branch model from an already existing node-breaker model, also called topology processing.

CIM supports both bus-branch and node-breaker modelling but lacks detail on how to do it resulting in several different ways.

A power flow performs the calculation on network with branch impedances and injections using Ohm's and Kirchhoff's laws. The branches and injections are connected at "fictive power flow busses", hence the term "bus branch model". In CIM the "fictive power flow busses" are represented by cim:TopologicalNodes (TN).

Closed switches represent a zero impedance which a power flow calculation doesn't handle easily. There are two approaches:

1. Represent closed switches with an impedance small enough not to impact the solution but big enough to enable power flow convergence. This have consequences:
   - May result in numerical instability when solving power flow.
   - Result in large matrices which degraded performance and increased need for memory space.

2. Remove all closed switches and merge the switch connection points called topology processing. This has consequences as well:
   - Topology processing is needed.
   - Mapping issues between the node breaker and bus branch models, e.g. transferring the power flow solution back to the objects replaced in topology processing.

Node-breaker modelling with CIM is supported by the class cim:ConnectivityNode (CN). To summarise CNs and TNs according to approach 2 above:

- CNs describe the as built network with or without switches.
- TNs describe the fictive power flow busses and are computed by topology processing when processing the switches.

Figure 28 show what an architecture supporting topology processing.

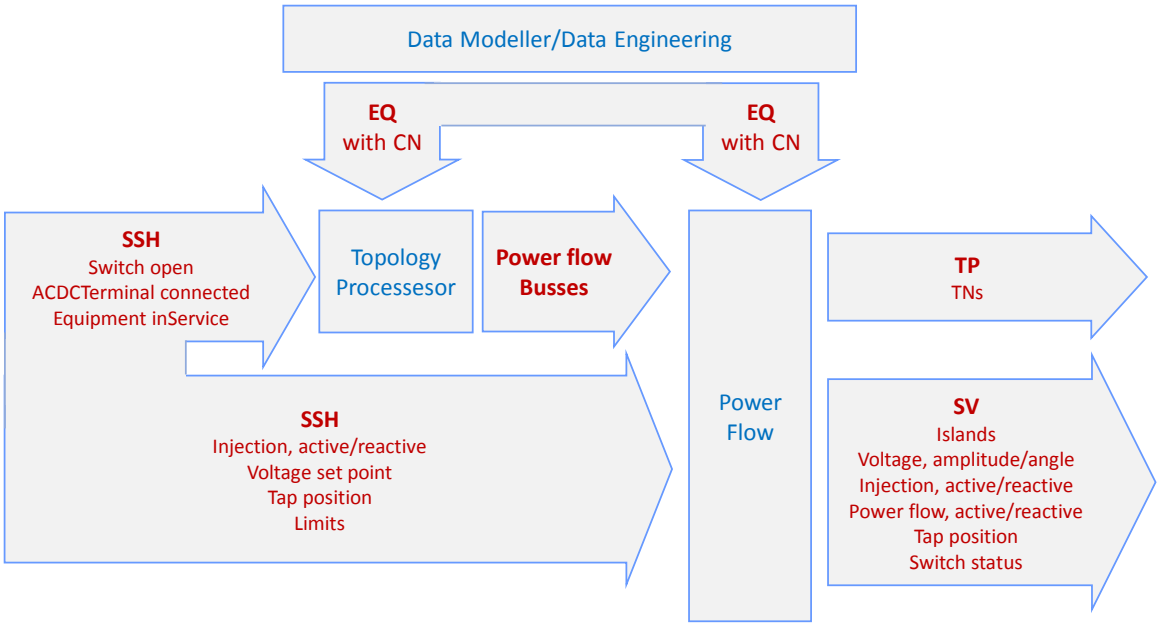



**Figure 28 Architecture with topology processing**

The as built CNs comes from a Data Modeller or Data Engineering tool where they are maintained. The CNs and all equipment connected to them are feed into the Topology Processor that create the TNs, hence the TNs are a result.

To summarize the discussion there are two types of models, node-breaker and bus-branch, and the as built model can be described with CNs or TNs. This give four possible combinations

A. A bus-branch model with the “as built model” using CNs.
B. A bus-branch model with the “as built model” using TNs.
C. A node-breaker model with the “as built model” using CNs.
D. A node-breaker model with the “as built model” using TNs

Figure 29 summarize the cases with recommended usage.

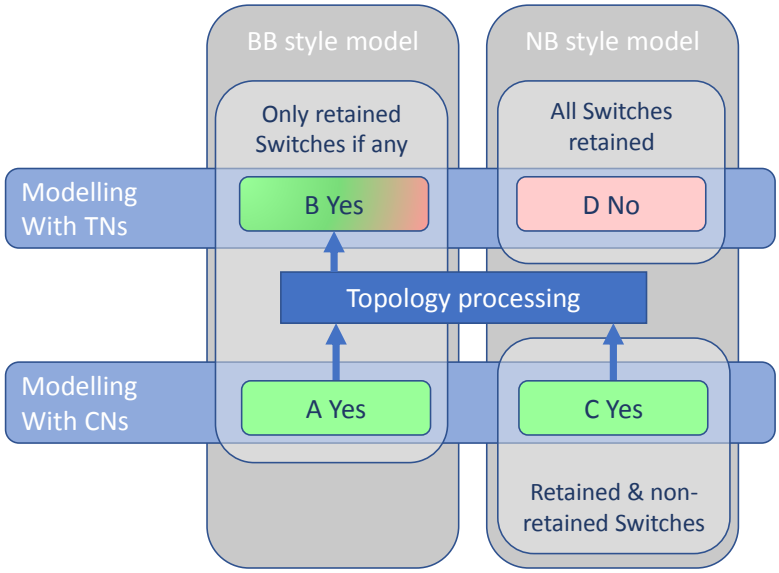



**Figure 29 As built modelling options using CNs or TNs**

Case A use the architecture described in Figure 34 and create the TNs by topology processing.

Case B will use TNs for the as built model and will stick to them for the result. Long term moves to case A.

Case C use the architecture described in Figure 34 and create the TNs by topology processing.

Case D is odd in that TNs are used to describe connectivity between switches in the as built model. No topology processing is done, and low impedance branches are used to represent the closed switches. It shall not be used. This case was not foreseen when CIM was created hence statements that ban it has unfortunately not been included in CIM.

A consequence of case B modelling is that voltages at open ended branches cannot be reported by CIMXML SV files, see Figure 30.

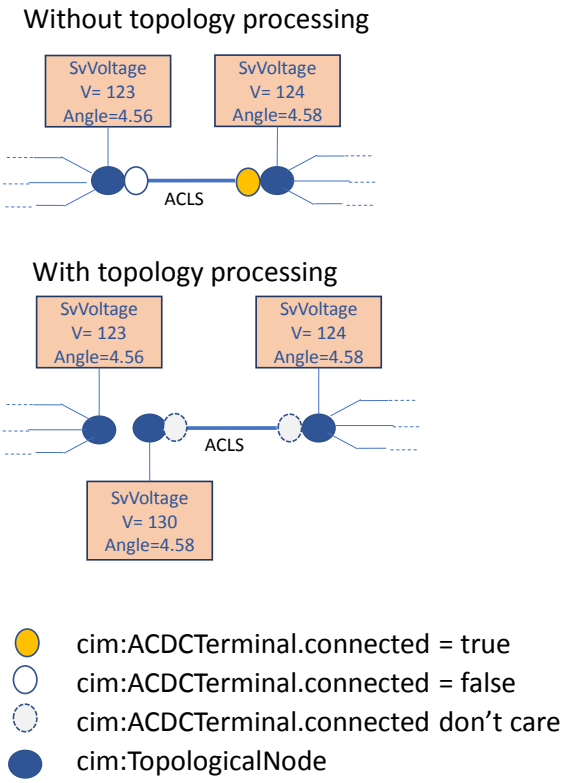



**Figure 30 Open ended branch in bus-branch model**

The TNs for an open-ended branch in case B where no topology processing is made is shown at the top. The TNs in the as built model isn't updated and no TN to describe the left open end is available, hence the voltage at the open end cannot be reported.

The TNs for an open-ended branch in case A are shown in the middle. All TNs calculated by the topology processor and the open-ended branch has got a TN where the voltage is reported.

A common solution to this in bus-branch tools is not to allow open ended branches.

A solution that avoids the issues discussed in this section is to use CNs for the as built model (case A and C in Figure 29) which means that the Data Modeller or Data Engineering tool is using CNs as in Figure 28. TNs are not used in the as built model but are created by topology processing. This has many benefits:

- A model that starts as a bus-branch without switches can be gradually refined by adding switch details. There are already several examples of IGMs where bus-branch and node-breaker styles are mixed in the same model.
- Network extensions are initially created as bus-branch models for study purposes and once the equipment is designed and built the original bus-branch model is extended with the detailed as built design.
- The issue with open ended branches disappears as the needed dead end TNs are created by topology processing.
- The boundary can be simplified and only contain CNs. TNs and the CIMXML TP boundary file is no longer needed which simplifies the assembly process.

## 10.6 Topology processing

The purpose of the Topology Processing function shown in Figure 28 is to create the cim:TopologicalNode(s) (TN) from the cim:ConnectivityNode(s) (CN) connected by closed (cim:Switch.open=false) and not retained (cim:Switch.retained=false) switches. Reason is to create a bus branch model where the closed switches, that represent a short circuit has been removed, as input to the power flow calculation according to discussed in section 10.5.

Switches that are retained (cim:Switch.retained=true) are treated as branches and are kept which means that their two ends (cim:Teminal(s)) shall always be connected to two different TNs. If this is not the case after topology processing, there may be another path with closed and non-retained switches between the two TNs. This is a modelling error and the IGM need to be corrected.

For CGMs TNs are used as input as shown in Figure 8. This means that the TNs shall be used as is and no topology processing be made. A merging tool may still do topology processing for NB IGMs included in the CGM to check that the provided TNs are correct. Despite any errors found the provided TNs shall still be used in the power flow calculation.

Creating TNs from CNs and closed switches is straight forward and can be summarized by the following graph search process:

1. Select an arbitrary CN and create a TN for the selected CN.
    a) Stop when all CNs have been visited.
2. For the selected CN
    b) Collect all neighbouring CNs connected via closed switches and add to CNs to be processed. Link all switches connected to the CN also to the TN (create reference cim:Terminal.TopologicalNode, see also section 10.8).
    c) Collect all branches and injections and link them to the TN (create references cim:Terminal.TopologicalNode).
3. Abandon the selected CN and select a new CN from the set of CNs to be processed.
    d) If a CN was found continue at 2
    e) If no more CNs in the set to process continue at 1.

This algorithm creates TNs for all CNs that are connected via closed switches and each TN will have injecting equipment and branches connecting with other TNs. Not all IGMs create TNs like this, there are IGMs with the following deviations

- Shunts with open switches that are still connected to an energized TN but with the cim:ShuntCompensator.sections=0 and/or cim:ACDCTerminal.connected=false. See section 10.16 for a discussion of this.
- Not energized TNs with a seemingly ad hoc collection of equipment. If such TNs are kept not energized they will not disturb the power flow solution but if energized they may disturb the power flow solution.
- TNs with erroneously connected equipment and branches, in this case the power flow solution is impacted.

## 10.7 TopologicalIsland

A cim:TopologicalIsland consist of cim:TopologicalNodes (TN) that are connected by branches

with non-zero impedance. A power flow solves each cim:TopologicalIsland individually and if it converges considers the island to be either energized or deenergized.

cim:TopologicalIsland(s) show how power flow applications have grouped TNs in islands and solved them. Hence the islands are an important tool to analyse and compare power flow results with big differences. The following checks when comparing can then be made:

- Islands are the same with same TNs
- Convergence status are the same for the islands, i.e. power flow was run for all islands and if it converged or not.
- Synchronous areas in a CGMs are reported as separate cim:TopologicalIsland(s) simplifying inspection of the synchronous areas.

TNs in the same cim:TopologicalIsland shall be electrically connected. Hence all TNs in an island have the same electrical status. The statuses are:

- Energized – the power flow solved for the island and non-zero injections present.
- Deenergized – the power flow solved for the island and all injections are zero.
- Diverged – the power flow didn't solve the island.

The cim:IdentifiedObject.description text of a cim:TopologicalIsland shall contain one of the texts:

- "Converged" for a converged solution.
- "Diverged" for a diverged solution.
- No other texts are allowed.

The voltage and angle for TNs are set as follows:

- For an energized island the voltage and angle are non-zero except for the reference bus where the angle is zero.
- For a deenergized or diverged island all voltages and angles are zero. Information about divergence is given by the power flow convergence report.

In diverged islands the computed injections and power flows in cim:SvPowerFlow shall be zero. The rules in QoCDC [3] that check the resulting power flow in SV with the input conditions in SSH will report errors or warnings due to this. To properly handle diverged islands, rules that check the resulting power flow needs to be revised to only check energized islands where cim:SvVoltages have nonzero voltage. Annex 1 list the rules that need update to check for zero voltage.

Small (typically too small to create a meaningful solution, see also rule SmallTopologicalIsland in QoCDC [3]) and deenergized cim:TopologicalIsland(s) is typically the result of topology processing where a few connected CNs are isolated and with zero injected power, e.g.

- Disconnected busbar section
- Open switches connected to a common CN
- Injections not delivering any power.



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Such cim:TopologicalIsland(s) shall not be exported as they do not add any value and clutter a solution.

## 10.8 Terminal to TopologicalNode reference

The reference cim:Terminal.TopologicalNode has cardinality 1 (required) in the TP profile but wording in IEC TS 61970-600-1:2017 section E.17 allow for exclusion of cim:Terminal.TopologicalNode from the CIMXML TP file when switch details are reduced away.

This has the consequences that:

- The voltage and angle on each side of a cim:Switch cannot be reported as cim:Terminal(s) are missing cim:Terminal.TopologicalNode.
- If the cim:Terminal at a cim:Switch is controlled by a cim:RegulatingControl or cim:TapChangerControl, see section 10.10, the reference to the controlled TN is lost and control cannot be performed.

Hence all cim:Terminal(s) shall have cim:Terminal.TopologicalNode according to the cardinality.

## 10.9 Switch retained

The cim:Switch.retained flag tells a power flow calculation to compute the power flow through a closed cim:Switch. This may be implemented by keeping the cim:Switch as a low impedance branch in the bus-branch model as discussed in section 10.5 approach 1.

The purposes of having retained cim:Switches are:

- Computing the power flow through a closed cim:Switch, this is useful both in node-breaker and bus-branch models.
- Allowing bus splits in a bus-branch model without having to edit the connectivity. It is common to make bus couplers retained for this reason.

## 10.10 Control configuration

The CIM information model on control allow for several ways to configure instances. This section discusses the options and their consequences. As basis for the discussion the diagram in Figure 31 is used.



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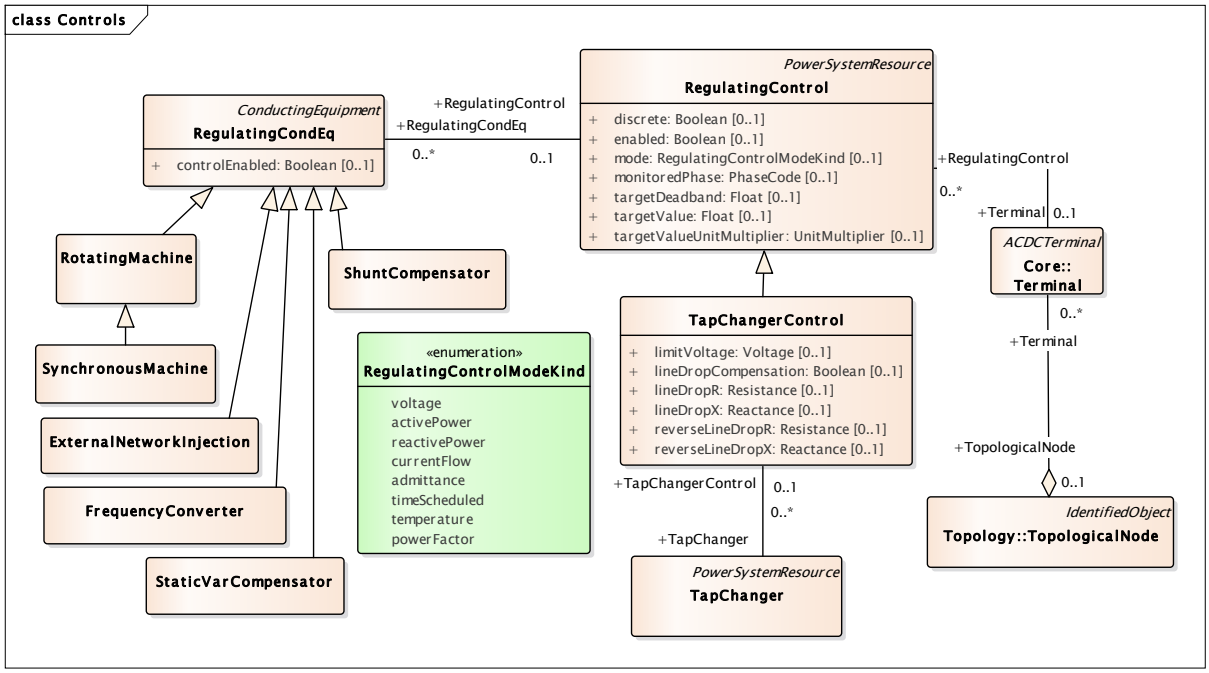



**Figure 31 CIM for control**

Figure 31 convey the following information:

- Controlling equipment is linked to a cim:RegulatingControl. The controlling equipment could be a subtype of
  - cim:RequlatingCondEq
  - cim:TapChanger
- A cim:RegulatingControl or cim:TapChangerControl may have many controlling equipment's.
- The quantity that cim:RegulatingControl controls is located at a cim:Terminal.
- A cim:Terminal may have many cim:RegulatingControl(s) or cim:TapChangerControl(s).
- A cim:Terminal is linked to a cim:TopologicalNode.
- A cim:TopologicalNode is where a power flow calculation control the quantity.
- A cim:TopologicalNode may have many cim:Terminal(s).

This gives several ways to instantiate the classes in Figure 31 as shown in Figure 32.

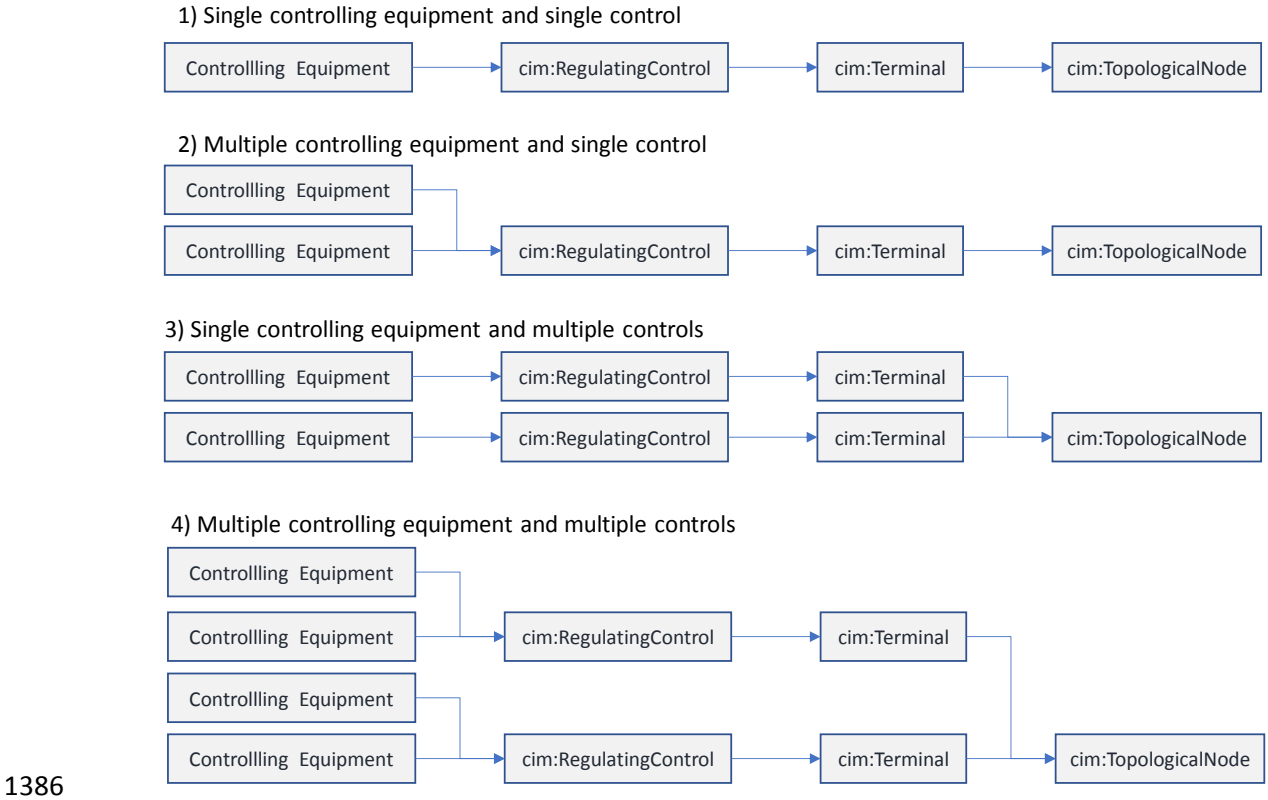



**Figure 32 Control configuration cases**

Cases 1 and 2 in Figure 32 are simple and without issues related to the control (cim:RegulatingControl or cim:TapChangerControl).

Cases 3 and 4 may have conflicts between cim:RegulatingControl.targetValue and cim:RegulatingControl.targetDeadBand between the different controls. The cim:RegulatingControl.targetValue(s) should all be the same but the cim:RegulatingControl.targetDeadBand(s) may differ depending on the controlling equipment, e.g. a tap changer may have a small dead band while a shunt may have a large.

In cases where multiple controlling equipment control the same TN the conclusions are:

- Use a single control to avoid target value conflicts.
- Use multiple controls only when different dead bands are needed.

In a node breaker model, it is possible to change the TN by switching. In a real substation breaker bays may allow for extensive reconfiguration of TNs, see Figure 33.



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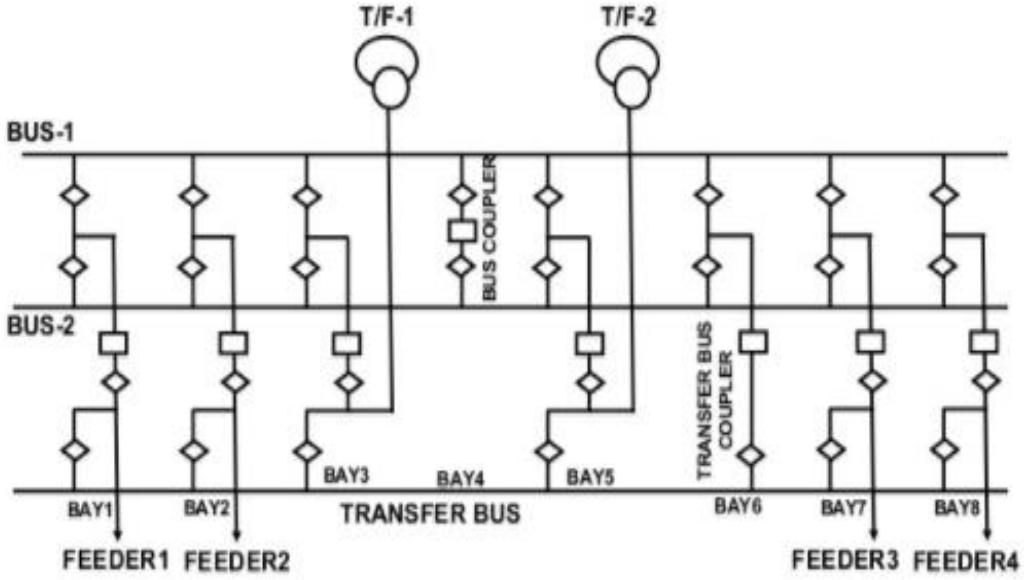



**Figure 33 Example node breaker substation**

The substation in Figure 33 has two main bus bars BUS-1 and BUS-2 and one transfer bus. The two machines T/F-1 and T/F-2 may be on different or the same TN depending of the switches. The control for machines T/F-1 and T/F-2 may be according to case 2) or case 3) in Figure 32. For the controlled cim:Terminal(s) this results in the following options:

1. With a single control (case 2) the controlled cim:Terminal could be at:
    a) one of the machines
    b) a switch
    c) one of the bus bars
2. With two controls (case 3) the controlled cim:Terminal could be at:
    a) one of the machines
    b) each of the machines
    c) a switch
    d) any two switches
    e) one of the bus bars
    f) any of two bus bars

With all options except 2b it is easy to lose the control by switching. In option 2b the control follows each machine to the TN where the machine is connected but if both machines are connected to the same TN the target values may instead conflict.

In a real substation it is common that the controlled point is located where the measurements are, e.g. at a breaker or bus bar. As concluded above this doesn't fit well with the need to make power flow calculations work with CIM data. Hence the CIM information model for control is insufficient and needs revision.

If the controlled cim:Terminal belong to a cim:Switch and a topology processor do not include cim:Terminal.TopologicalNode in the CIMXML TP file this is another reason for loss of control capability, refer to discussion in section 10.5.

## 10.11 Equivalents

### 10.11.1 EquivalentInjection

The class cim:EquivalentInjection has three different usages:

1. Represent boundary flows between IGMs, refer to section 9.4.
2. Equivalent representation of HVDC flows.
3. Reduced network parts in a model.

Case 1 is distinguished by the cim:EquivalentInjection(s) being connected to a boundary TN or CN.

Case 2 is distinguished by the cim:EquivalentInjection(s) being connected to a boundary TN or CN with the text “HVDC” in the beginning of the TN or CN cim:IdentifiedObject.description attribute.

Case 3 distinguished by the cim:EquivalentInjection(s) not meeting case 1 or case 2.

Case 1 and 2 require an assembly between EQ and EQBD CIMXML files to be determined.

### 10.11.2 EquivalentBranch

Cim:EquivalentBranch(es) may include transformers, hence they may span between different base voltages. But cim:EquivalentBranch(es) do not have the attribute phaseAngleClock so phase angle displacement cannot be described. Hence cim:EquivalentBranch(es) cannot be used to describe transformers with non-zero cim:PowerTransformerEnd.phaseAngleClock. In this case a cim:PowerTransformer must be used.

## 10.12 LoadResponseCharacteristic

RSCs shall ignore cim:LoadResponseCharacteristic instances with exponential dependency on voltage (cim:LoadResponseCharacteristic.exponentModel = true). The load given by cim:EnergyConsumer.p and cim:EnergyConsumer.q shall be used as is without using the equations for voltage dependency.

TSOs may use instances of cim:LoadResponseCharacteristic with exponential dependency on voltage. If used the load values (cim:EnergyConsumer.p and cim:EnergyConsumer.q) shall be specified at nominal voltage. If the power flow solution ends at another voltage than nominal the load will deviate from the nominal and the resulting load is given by cim:SvPowerFlow p and q.

## 10.13 Line taps and T-points

A transformer at a transmission line is sometimes modelled as an injection at the line, a “tap”. High voltage transmission lines typically don’t have taps, but lower voltage distribution feeders have.

Network models used in processes described in this document do not include low voltage feeders with taps. If a transmission line has a tap, switching equipment as well as a transformer is normally present, hence the tap shall be described being within a cim:Substation and not in a cim:Line.

A cim:Line may have a T-point with an additional cim:ACLineSegment branches off from the line. A T-point in a cim:Line has three cim:ACLineSegment(s) connected to a CN/TN. More than one T-point may exist in a cim:Line.



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## 10.14 Production in distribution networks

Production may be embedded in distribution networks, e.g. in case of renewable production. The distribution network typically connects to the transmission network at a transformer. The flow through the transformer will then have both a load and production component. If the flow is modelled as a load, e.g. a cim:ConformLoad, the expected load variation curve during a day will be distorted by the production component. To remedy this the load and production shall be modelled separately, i.e.

- The load shall be modelled by a cim:ConformLoad (housing) and/or cim:NonConformLoad (industrial) depending of the behaviour.
- The production shall be modelled by an aggregate cim:SyncronousMachine or cim:EquivalentInjection. Both have voltage control capability that may be utilized if the production has reactive resources.

## 10.15 StaticVarCompensator (SVC)

### 10.15.1 Overview

Reactive power output of a cim:StaticVarCompensator (SVC) is limited by the capacitive and inductive ratings representing the maximum and minimum susceptance values derived from reactive power output of capacitor and reactor banks. In normal operations these limits are not often reached.

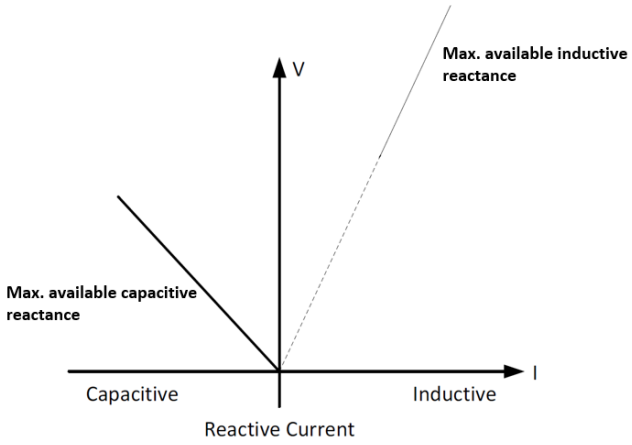



**Figure 34 Capability curve for a SVC**

The min/max reactive power of a SVC determined by:

$$Q_{min/max} = -\frac{V \times V}{rating}$$

Where:

- rating is the capacitiveRating or inductiveRating
- V = local voltage at the SVC. Even when in remote regulation, the local voltage shall be considered to retrieve the minimum or maximum local reactive output.
- Minus sign in equation according to:

- IEC 61970-600-2:2017 The value of the inductiveRating is negative, the value of the capacitiveRating is positive.
- IEC 61970-301:2016 positive current (I in Figure 34) indicates an inductive current as load sign convention is used.

### 10.15.2 Implementing Control

IEC 61970-301:2016 and IEC 61970-600-2:2017 do not specify if the local SVC control attributes shall be used or the cim:RegulatingControl attributes. As the cim:RegulatingControl describes the controlled point in the network (a cim:Terminal), cim:RegulatingControl shall be used and not the SVC local attributes:

- cim:StaticVarCompensator.sVCControlMode
- cim:StaticVarCompensator.voltageSetPoint

cim:StaticVarCompensator.q may be used as a reactive starting point in power flow. It shall not be used as targetvalue for a reactive regulation.

cim:RegulatingControl.enabled defines whether the regulation is enabled. When the attribute is false, it means the regulation is off and the reactive output of the SVC shall be zero Mvar.

An SVC may operate in fixed Mvar output mode or in voltage control mode. This is defined by the RegulatingControl.mode which for an SVC could be voltage or reactivePower.

### 10.15.3 Voltage Control

When the cim:StaticVarCompensator.slope is zero, the SVC will inject or absorb reactive power to reach the voltage target as long as capacitiveRating or inductiveRating limits are not reached.

When the cim:StaticVarCompensator.slope is positive, the reactive output of the SVC is defined by:

$$Q = \frac{V_{bus} - V_{ref}}{slope}$$

Where:

- cim:StaticVarCompensator.slope is positive based on the load sign convention.
- Vbus is the voltage at the controlled point determined by cim:RegulatingControl.Terminal which may be local or remote.
- Vref is the target value given by cim:RegulatingControl.targetValue.

### 10.15.4 Reactive power control

According to IEC 61970-301:2016 an SVC in reactive power control keeps the susceptance constant. This is not correct and in reality, the reactive power flow at controlled point is kept constant by varying the susceptance so the SVC absorb or inject reactive power to reach the reactive target value at the controlled point.

The reactive power control could apply both to a local or to a remote-controlled point. In case of remote-controlled point, load-flow tools should assess the direction to change reactive power flow on the SVC.

## 10.16 Shunt compensators

Shunt compensators are often used in power flow to control voltage. It is also common they are used together with equipment with continuous control capability as synchronous machines and static var compensators with the purpose to keep continuous resources in the middle of their control range. A way to solve this in a power flow is to let the shunts control the reactive power flow from the continuous resource so the flow is close to zero. This can be done with the configuration shown in Figure 35.

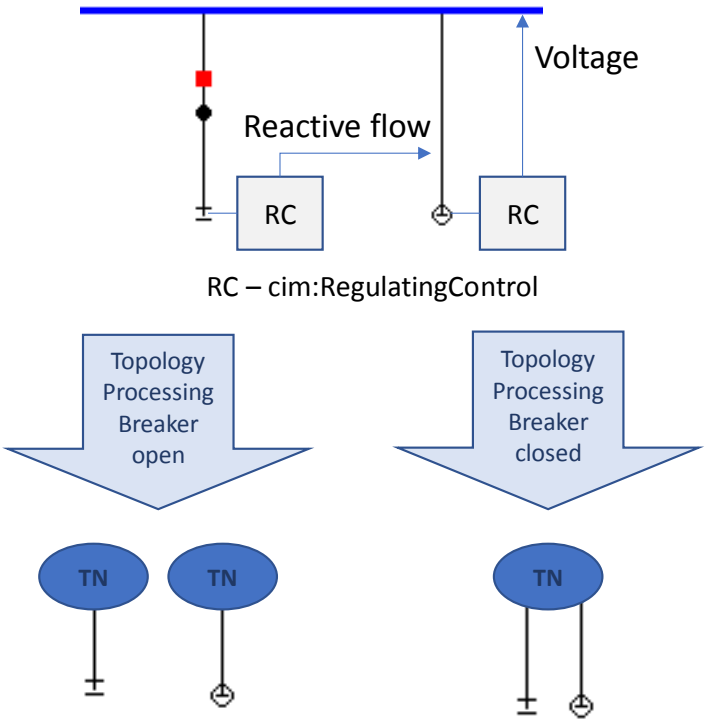



**Figure 35 Voltage control with shunts and continuous resource**

When the synchronous machine in Figure 35 delivers or consumes a large amount of reactive power to maintain the voltage the shunt will change its capacitance or reactance to lessen the reactive power from the synchronous machine.

If the reactive power is controlled by switching the shunt, a topology processor will create a separate TN if the shunt is disconnected as shown to the lower left in Figure 35. To get the shunt connected back to the TN with the synchronous machine requires closing the switch and repeated topology processing. As the CGM power flow is TN based (see Figure 8), topology processing isn't used as discussed in section 10.6. Instead the following solution is to be used:

- Switches shall not be used to control the shunt impedance and be closed when the shunt is in service, e.g. the shunt breaker in Figure 35 is only used when the shunt is taken out of service.
- Use the cim:ShuntCompensator.sections to control the impedance where cim:ShuntCompensator.sections=0 means zero admittance. For linear shunt this means multiplying with the per section admittance results in zero impedance. For nonlinear shunts

the cim:NonlinearShuntCompensatorPoint.sectionNumber=0 shall have the corresponding admittance values equal zero.

## 10.17 Series compensators

Series compensators may appear both in a substation and somewhere along a transmission line. Modelling a series compensator along a line can be made in two ways:

1. Implicit by adding the series reactance to the cim:ACLineSegment impedances.
2. Explicit by adding the series compensator as a cim:SeriesCompensator together with other equipment that is typically present, e.g. breakers and disconnectors. This require a cim:Substation at the location of the cim:SeriesCompensator.

## 10.18 Power Transformers

For two winding transformers both series and shunt elements are reactive which means that the series element shall be positive and the shunt element negative. But for equivalenced or three winding transformers a shunt element (cim:PowerTransformerEnd.b) could be positive. If a cim:PowerTransfomer is included as an equivalenced cim:PowerTransfomer the cim:Equipment.aggregate flag shall be true and shunt elements could be positive.

The electrical equivalent of a two winding transformer is an L-model, see Figure 36.

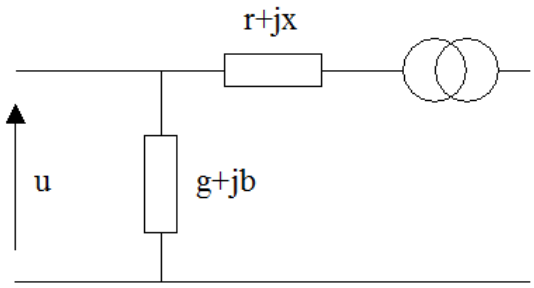



**Figure 36 Two Winding Power Transformer L-model**

As can be seen from Figure 36 the model has one shunt and one series element. But the CIM information model has two cim:PowerTransformerEnd(s) that can store one shunt and one series element each. To avoid ambiguity it has been decided to store the impedances at the high voltage cim:PowerTransformerEnd which is also end number 1. The other end shall have zero impedance values.

The equivalent model for a three winding transformer is L-elements in a Y-configuration where the shunt elements are at the cim:PowerTransformer cim:Terminal(s). In this case the three pairs of impedance elements are stored in each of the three cim:PowerTransformerEnd(s). As the shunt impedance is typically measured at one of the transformer windings, the shunt impedance should be stored in this cim:PowerTransformerEnd rather than being distributed over all three, see also Figure 37.

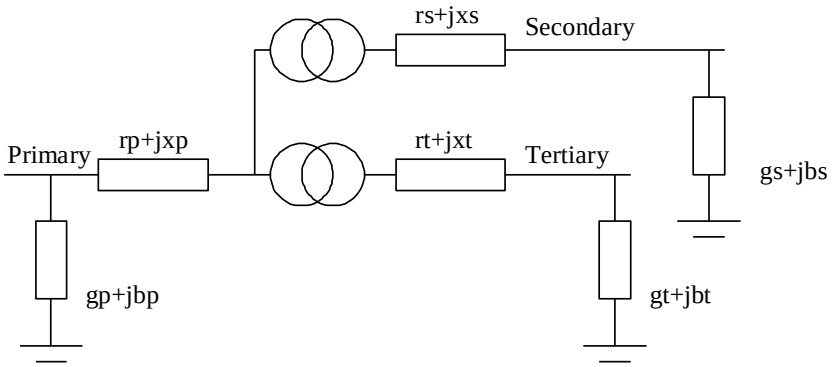



**Figure 37 Three Winding Power Transformer L-model**

## 10.19 Tap changer modelling

## 10.20 This section is intentionally empty.Reactive limits

The reactive limits of equipment that can supply reactive power (e.g. cim:SynchronousMachine, cim:EquivalentInjection) is either described by a pair of min/max limit attributes or a reactive capability curve. If a reactive capability curve is present any provided min/max limit attributes shall be ignored.

## 10.21 Attributes not used

There are several attributes that are not used but being exchanged in IGMs. Some are defined in the IEC TS 61970-600-1/2:2017 profile and some in canonical CIM, see Table 6.

All classes, roles and attributes not described in a profile shall be ignored and reported for diagnostics.

1596 **Table 6 Not applicable classes or attributes**

| Attribute | Defined in | Comment |
|---|---|---|
| EquivalentBranch.r21<br>EquivalentBranch.x21 | CGMES | The attributes EquivalentBranch.r21 and EquivalentBranch.x21 shall be symmetric, i.e. have the same values as EquivalentBranch.r and EquivalentBranch.x. For this reason, they are not needed and shall not be used. |
| PhaseTapChangerNonLinear.xMin | CGMES | This attribute is redundant with PhaseTapChangerNonLinear.xMax and shall not be used. |
| GeneratingUnit.initialP | CGMES | This attribute can be used as a starting point to power flow when no other data is available. The SSH file contain all the needed inputs. |
| VoltageLevel.highVoltageLimit<br>VoltageLevel.lowVoltageLimit | CGMES | The limits used are from OperationalLimits, these attributes are not used. |
| EnergyConsumer.pfixed<br>EnergyConsumer.qfixed<br>EnergyConsumer.pfixedPct<br>EnergyConsumer.qfixedPct | CGMES | This attribute can be used to create a starting point to power flow when no other data is available. The SSH file contain all the needed inputs. |

1597

1598 **Table 6(continuing) Not applicable classes or attributes**

| Attribute | Defined in | Comment |
|---|---|---|
| SeriesCompensator.varistorPresent<br>SeriesCompensator.varistorRatedCurrent<br>SeriesCompensator.varistorThresholdVoltage | CGMES | These attributes are used for over voltage protection and not applicable. |
| ShuntCompensator.switchOnCount<br>ShuntCompensator.switchOnDate | CGMES | These attributes are used for asset health supervision and are not applicable. |
| Meas package attributes<br>Accumulator<br>AccumulatorLimit<br>AccumulatorLimitSet<br>AccumulatorReset<br>AccumulatorValue<br>Analog<br>AnalogLimit<br>AnalogLimitSet<br>AnalogValue<br>Command<br>Discrete<br>DiscreteValue<br>MeasurementValueQuality<br>MeasurementValueSource<br>RaiseLowerCommand<br>SetPoint<br>StringMeasurement<br>StringMeasurementValue<br>ValueAliasSet<br>ValueToAlias | CGMES | These attributes are used to describe measurements, e.g. in a SCADA/EMS, and are not applicable. |
| BusNameMarker | CGMES | This class is used in node breaker models and topology processing to generate cim:TopologicalNodes with the same name and ID. Currently not applicable but may be in the future. |

1599 **Table 6(continuing) Not applicable classes or attributes**

| Attribute | Defined in | Comment |
|---|---|---|
| Pattern data<br>Season<br>DayType<br>ConformLoadSchedule<br>NonConformLoadSchedule<br>RegulationSchedule<br>SwitchSchedule<br>TapSchedule | CGMES | These classes are used to set up power flow inputs, e.g. loads and switch position. Not applicable as this information is already in SSH.<br>In the future they could be used to support creation of SSH. |
| Junction | CGMES | This class is currently used in the boundary in case location information is needed. In CGMES 3.0 the class is no longer needed as locations are managed by the new class BoundaryPoint. The class shall not be used. |
| Real time operational information<br>OperatingParticipant<br>OperatingShare<br>ReportingGroup | CGMES | The use of these classes is not well defined. Some use them as a way to determine joint operational responsibilities, not applicable. |
| BasePower | CIM | Used for PU scaling and is not needed as all data is in engineering units, not applicable. |
| Flow supervision<br>BranchGroup<br>BranchGroupTerminal | CIM | Used to supervise power flow in a set of branches typically operating in parallel. Currently not applicable but may be in the future when SIPS is modelled. |
| PSRType | CIM | The class is intended for custom extensions, not applicable. |
| BranchLimitPriority | Not in CIM | BranchLimitPriority seem to be a custom extension but given the cim: name space prefix which is illegal. |

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1601

# Annex 1 – Rules checking SvPowerFlow that also need to be checked for SvVoltage

DisconnectedTerminal

GenActivePowerInfeedDiffW

GenActivePowerInfeedDiffE

GenReactivePowerInfeedDiffW

DERActivePowerInfeedDiffW

DERActivePowerInfeedDiffE

DERReactivePowerInfeedDiffW

LoadActivePowerInfeedDiffW

LoadActivePowerInfeedDiffE

LoadReactivePowerInfeedDiffW

ENIActivePowerInfeedDiffW

ENIActivePowerInfeedDiffE

ENIReactivePowerInfeedDiffW

EIActivePowerInfeedDiffW

EIActivePowerInfeedDiffE

EIReactivePowerInfeedDiffW

NetInterchange1

NetInterchange2

ShuntQ

ACScheduleMatch1

ACScheduleMatch2

HVDCScheduleMatch1

HVDCScheduleMatch2

CGMCongestion

CGMTieFlowImbalance



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# Annex 2 – References to attributes and roles from QoCDC rules

The appendix contains tables listing references from the rules to the information model.

Restrictions on the XML format and syntax appears as level 2 rules, see Table 7.

**Table 7 XML format and syntax rules**

| Level 2 Rule | Schema deviation |
|---|---|
| AttributeAndRoleValues | Format of role and attribute XML elements |
| ValidResourceValue | Format of resource references |
| ValidAboutValue | Format of ID attribute rdf:about |
| ValidIDValue | Format of ID attribute rdf:ID |
| DecimalComma | Format of decimal separator |
| NotaNumber | Format of numeric values |



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1635 The level 3 rules in Table 8 describe allowed values for codes or max length of names.

1636 **Table 8 Name length rules**

| Level 3 Rule | Restricted attribute value |
|---|---|
| NameLength | cim:IdentifiedObject.name |
| DescriptionLength | cim:IdentifiedObject.description |
| EICLength | entsoe:IdentifiedObject.energyIdentCodeEic |
| ShortNameLength | entsoe:IdentifiedObject.shortName |
| CNFromEndIsoCode | entsoe:ConnectivityNode.fromEndIsoCode |
| TNFromEndIsoCode | entsoe:TopologicalNode.fromEndIsoCode |
| CNToEndIsoCode | entsoe:ConnectivityNode.toEndIsoCode |
| TNToEndIsoCode | entsoe:TopologicalNode.toEndIsoCode |
| CNFromEndNameLength | entsoe:ConnectivityNode.fromEndName |
| TNFromEndNameLength | entsoe:TopologicalNode.fromEndName |
| CNToEndNameLength | entsoe:ConnectivityNode.toEndName |
| TNToEndNameLength | entsoe:TopologicalNode.toEndName |
| CNFromEndNameTsoLength | entsoe:ConnectivityNode.fromEndNameTso |
| TNFromEndNameTsoLength | entsoe:TopologicalNode.fromEndNameTso |
| CNToEndNameTsoLength | entsoe:ConnectivityNode.toEndNameTso |
| TNToEndNameTsoLength | entsoe:TopologicalNode.toEndNameTso |
| DCEquipmentContainerMapping | entsoe:IdentifiedObject.energyIdentCodeEic |

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1638 The level 3 rules in Table 9 describe restricted attribute values within a single CIMXML file.

1639 **Table 9 Restricted attribute values**

| Level 3 Rule | Restricted attribute value |
|---|---|
| ShuntCompensatorSensitivity | cim:ShuntCompensator.voltageSensitivity |
| GeneratingUnitNominalP | cim:GeneratingUnit.nominalP |
| NominalVoltage | cim:BaseVoltage.nominalVoltage |
| TerminalSeqNum | Presence of cim:Terminal.sequenceNumber |
| TerminalSeqNumOrder | cim:Terminal.sequenceNumber |
| LRCExponentModel | Restrictions on<br>cim:LoadResponseCharacteristic.pVoltageExponent,<br>cim:LoadResponseCharacteristic.qVoltageExponent<br>Based on<br>cim:LoadResponseCharacteristic.exponentModel, |
| LCRCoefficientModel | Restrictions on<br>cim:LoadResponseCharacteristic.pConstantImpedance<br>cim:LoadResponseCharacteristic.pConstantCurrent<br>cim:LoadResponseCharacteristic.pConstantPower<br>cim:LoadResponseCharacteristic.qConstantImpedance<br>cim:LoadResponseCharacteristic.qConstantCurrent<br>cim:LoadResponseCharacteristic.qConstantPower.<br>Based on<br>cim:LoadResponseCharacteristic.exponentModel. |
| LCRCoefficientParameters | Restrictions on sums of<br>cim:LoadResponseCharacteristic.pConstantImpedance<br>cim:LoadResponseCharacteristic.pConstantCurrent<br>cim:LoadResponseCharacteristic.pConstantPower<br>cim:LoadResponseCharacteristic.qConstantImpedance<br>cim:LoadResponseCharacteristic.qConstantCurrent<br>cim:LoadResponseCharacteristic.qConstantPower.<br>Based on<br>cim:LoadResponseCharacteristic.exponentModel. |
| MeasType | cim:Measurement.measurementType |
| MeasUnit | cim:Measurement.unitSymbol |
| TargetDB | cim:RegulatingControl.targetDeadband |

1640

1641 **Table 9 (continuing) Restricted attribute values**

| Level 3 Rule | Restricted attribute value |
|---|---|
| OperationalLimitValue | cim:VoltageLimit.value,<br>cim:CurrentLimit.value,<br>cim:ActivePowerLimit.value,<br>cim:ApparentPowerLimit.value |
| AcceptableDuration | cim:OperationalLimitType.acceptableDuration |
| PATL4 | cim:CurrentLimit.value,<br>cim:ActivePowerLimit.value,<br>cim:ApparentPowerLimit.value |
| PATL5 | cim:VoltageLimit.value, |
| EnergySourceVoltage | cim:EnergySource.voltageMagnitude,<br>cim:EnergySource.voltageAngle |
| ControlModeCompatibility | cim:RegulatingControl.mode |
| ACLineSegmentR | cim:ACLineSegment.r |
| ACLineSegmentX | cim:ACLineSegment.x |
| SeriesCompensatorX | cim:SeriesCompensator.x |
| EquivalentBranchX | cim:EquivalentBranch.x, cim:EquivalentBranch.x21 |
| PowerTransformerEndR | cim:PowerTransformerEnd.r |
| PowerTransformerEndRatedU | cim:PowerTransformerEnd.ratedU |
| PowerTransformerEndX | cim:PowerTransformerEnd.x |
| LinearShuntCompensatorG | cim:LinearShuntCompensator.gPerSection |
| ShuntCompensatorSections | cim:ShuntCompensator.normalSections |
| SVCRatings | cim:StaticVarCompensator.capacitiveRating,<br>cim:StaticVarCompensator.inductiveRating. |
| SVCSlope | cim:StaticVarCompensator.slope |
| GeneratingUnitLimits | cim:GeneratingUnit.maxOperatingP,<br>cim:GeneratingUnit.minOperatingP. |
| GeneratingUnitMaxPGen | cim:GeneratingUnit.maxOperatingP |
| SMQLimits1 | cim:SynchronousMachine.maxQ,<br>cim:SynchronousMachine.minQ. |

1642

1643 **Table 9 (continuing) Restricted attribute values**

| Level 3 Rule | Restricted attribute value |
|---|---|
| RatedS | cim:RotatingMachine.ratedS,<br>cim:PowerTransformerEnd.ratedS |
| SMQLimits3 | cim:GeneratingUnit.maxOperatingP,<br>cim:SynchronousMachine.maxQ,<br>cim:GeneratingUnit.minOperatingP,<br>cim:SynchronousMachine.minQ,<br>cim:RotatingMachine.ratedS |
| CurveStyle | cim:Curve.curveStyle |
| RCCYValues | cim:CurveData.y2value, cim:CurveData.y1value |
| CurveXValues | cim:CurveData.y2value, cim:CurveData.y1value,<br>cim:CurveData.xvalue between different cim:CurveData. |
| RCCXValues3 | CurveData.xvalue, GeneratingUnit.minOperatingP,<br>GeneratingUnit.maxOperatingP |
| RCCXValues4 | cim:CurveData.xvalue, cim:CurveData.y1value,<br>cim:CurveData.y2value, cim:RotatingMachine.ratedS |
| VSCYValues | cim:CurveData.y2value, cim:CurveData.y1value |
| PhaseCodeGround | cim:Terminal.phases |
| DCLineSegmentR | cim:DCLineSegment.resistance,<br>cim:PerLengthDCLineParameter.resistance |
| ConverterLosses | cim:ACDCConverter.idleLoss,<br>cim:ACDCConverter.switchingLoss<br>cim:ACDCConverter.resistiveLoss |

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1645 The level 3 rules in Table 10 describe the containments for equipment in EQ and EQBD CIMXML
1646 files.

1647 **Table 10 Containment rules local within a CIMXML file**

| Level 3 Rule | Contained class |
|---|---|
| GenerationContainment | cim:HydroPump and cim:GeneratingUnit |
| PTContainment | cim:PowerTransformer |
| SwitchContainment | cim:Switch |
| SCContainment | cim:SeriesCompensator |
| InjectionContainment | cim:EnergyConsumer, cim:RotatingMachine, cim:ShuntCompensator, cim:EnergySource, cim:EquivalentShunt, cim:ExternalNetworkInjection and cim:StaticVarCompensator. |
| BusbarSectionContainment | cim:BusbarSection |
| EFCContainment | cim:EarthFaultCompensator |
| EquivalentInjectionContainment | cim:EquivalentInjection |
| JunctionContainment | cim:Junction |
| CNContainment | cim:ConnectivityNode |
| ACDCConvContainment | cim:CsConverter, cim:VsConverter |
| DCEQContainment | cim:DCSeriesDevice, cim:DCShunt, cim:DCBusbar, cim:DCGround, cim:DCChopper, cim:DCSwitch, cim:DCBreaker, cim:DCDisconnector |
| DCNodeContainment | cim:DCNode |

1648

1649 The level 3 rules in Table 11 describe restrictions on relations local within a CIMXML file.

1650 **Table 11 Rules on restricted relations**

| Level 3 Rule | Restricted relation |
|---|---|
| CNTerminals | cim:Terminal.ConnectivityNode.<br>A CN shall not be isolated or dead end. |
| CEBaseVoltage | A cim:ConductingEquipment shall have a cim:BaseVoltage.<br>Multiple paths from cim:ConductingEquipment to cim:BaseVoltage are evaluated. |
| TerminalCount1 | cim:Terminal.ConductingEquipment<br>Classes that shall have a single cim:Terminal |
| TerminalCount2 | cim:Terminal.ConductingEquipment<br>Classes that shall have a two cim:Terminals |
| PTTerminalConsistency | Consistency between roles<br>cim:PowerTransformerEnd.PowerTransformer and cim:TransformerEnd.Terminal |
| MCFirst | cim:MutualCoupling.First_Terminal must refer to a cim:Terminal at a cim:ACLineSegement cim:Terminal. |
| MCSecond | cim:MutualCoupling.First_Terminal must refer to a cim:Terminal at a cim:ACLineSegement cim:Terminal. |
| MeasTerminal | Consistency between roles cim:Measurement.Terminal and cim:Measurement.PowerSystemResource. |
| OperationalLimitSetAtTerminal | cim:OperationalLimitSet.Terminal |
| PATL1 | cim:OperationalLimitSet.Terminal for cim:Terminals at cim:ACLineSegment and cim:SeriesCompensator. |
| PATL2 | cim:OperationalLimitSet.Terminal for cim:Terminals at cim:PowerTransformer. |
| PATL3 | cim:OperationalLimit.OpertionalLimiSet for classes cim:ApparentPowerLimit, cim:ActivePowerLimit, cim:CurrentLimit |
| ReactiveControlAtBus | cim:RegulatingControl.Terminal |
| SynchronousCondenser | cim:GeneratingUnit.SynchronousMachine, cim:SynchronousMachine.type |
| SMQLimits2 | cim:SynchronousMachine.InitialReactiveCapabilityCurve |
| RCandTCCcontrollingObjects | cim:RegulatingCondEq.RegulatingControl, cim:TapChanger.TapChangerControl |



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1651 The level 3 rules in Table 12 describe restrictions on number of instances for a class in a
1652 CIMXML file.

1653 **Table 12 Rules on restricted classes**

| Level 3 Rule | Restricted class |
|---|---|
| NumberOfSubstations | Number of cim:Substations in an IGM. |
| InstancesOfGeneralClass | The most specific and detailed class shall be instantiated.<br>cim:EnergyConsumer |
| CNRequiredInEQOperations | cim:ConnectivityNode |
| CATieFlow | cim:TieFlow |
| RCCXValues2 | cim:CurveData |
| VSCXValues | cim:CurveData |
| ControlAreaInstance | cim:ControlArea, cim:ControlArea.type,<br>entsoe:IdentifiedObject.energyIdentCodeEic |

1654

1655 The level 5 rules in Table 13 describe restricted attribute values where the restriction depends
1656 on multiple CIMXML files.



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1657 **Table 13 Restricted attribute values depending on multiple CIMXML files**

| Level 5 Rule | Restricted attribute value |
|---|---|
| GeographicalRegionsBD | cim:IdentifiedObject.name |
| SVCVoltage | cim:RequlatingControl.targetValue,<br>cim:RequlatingControl.enabled,<br>cim:RequlatingControl.mode,<br>cim:RequlatingControl.Terminal |
| TapChangerNeutralU | cim:TapChanger.neutralU,<br>cim:PowerTransformer.ratedU |
| EquivalentInjectionControl Enabled | cim:EquivalentInjection.regulationCapability,<br>cim:IdentifiedObject.description |
| NoLTCTapChangerControl | cim:TapChanger.ltcFlag |
| DisconnectedTerminal | cim:Terminal.connected, cim:SvPowerFlow.p,<br>cim:SvPowerFlow.q |
| SwitchOpenVsConnected | cim:ACDCTerminal.connected, cim:Switch.open |
| TapChangerTargetRange | cim:RegulatingControl.targetValue |
| IDuniqueness | rdf:ID, rdf:about |

1658

1659 The level 5 rules in Table 14 describe allowed containments for equipment in EQ and EQBD
1660 CIMXML files. Equipment in an EQ CIMXML file may be contained by containers in the EQBD.

1661 **Table 14 Containment rules that may involve multiple CIMXML files**

| Level 5 Rule | Contained class |
|---|---|
| LineContainment | cim:ACLineSegment, possibly in the boundary. |
| DCLineContainment | cim:DCLineSegment |

1662



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1663 The level 5 rules in Table 15 describe restrictions on relations across CIMXML files.

1664 **Table 15 Rules on restricted relations across CIMXML files**

| Level 5 Rule | Restricted relation |
|---|---|
| ControlLinkedToTopology | cim:RegulatingControl.Terminal,<br>cim:Terminal.TopologicalNode |
| BranchBaseVoltage | cim:ConductingEquipment.BaseVoltage,<br>cim:TransformerEnd.BaseVoltage |
| SvPowerFlowBranchInstances | cim:Terminal.ConductingEquipment,<br>cim:Terminal.TopologicalNode |
| SlackNode | cim:TopologicalIsland.AngleRefTopologicalNode |
| SwitchTerminals | cim:Terminal.ConductingEquipment,<br>cim:Terminal.ConnectivityNode |
| SwitchVL | cim:Terminal.ConductingEquipment,<br>cim:Terminal.ConnectivityNode,<br>cim:Terminal.TopologicalNode,<br>cim:TopologicalNode.BaseVoltage,<br>cim:Equipment.EquipmentContainer,<br>cim:VoltageLevel.BaseVoltage |
| SwitchTN1 | cim:Terminal.ConductingEquipment,<br>cim:Terminal.TopologicalNode, |
| ControlOfAnotherIsland | cim:RequlatingCondEq.RegulatingControl,<br>cim:TapChanger.TapChangerControl,<br>cim:RegulatingControl.Terminal,<br>cim:Terminal.TopologicalNode,<br>cim:TopologicalIsland.TopologicalNode |

1665

1666 The level 5 rules in Table 16 describe restrictions on number of instances for a class in
1667 depending of multiple CIMXML files.

1668 **Table 16 Rules on restricted classes across CIMXML files**

| Level 5 Rule | Restricted class |
|---|---|
| GeographicalRegion | cim:GeographicalRegion |
| BaseVoltageNotInBoundary | cim:BaseVoltage |
| SvTapStepInstances | cim:SvTapStep |
| SvPowerFlowInstances | cim:SvPowerFlow |
| SvPowerFlowBranchInstances2 | cim:SvPowerFlow |
| TopologicalIslandInstance | cim:TopologicalIsland |
| SmallTopologicalIsland | cim:TopologicalIsland |
| ParticipatingGeneratingUnit | cim:GeneratingUnit |

1669

1670 The level 6 rules in Table 17 describe restricted attribute values within an assembled model.

1671 **Table 17 Restricted attribute values in an assembled model**

| Level 6 Rule | Restricted attribute value |
|---|---|
| SCSections | cim:ShuntCompensator.sections |
| GenActivePowerInfeedLim | cim:RotatingMachine.p |
| GenActivePowerInfeedDiffW | cim:SvPowerFlow.p |
| GenActivePowerInfeedDiffE | cim:SvPowerFlow.p |
| GenReactivePowerInfeedDiffW | cim:SvPowerFlow.q |
| GenReactivePowerInfeedLim | cim:RotatingMachine.q |
| GenRCCPowerInfeed | cim:RotatingMachine.p, cim:RotatingMachine.q |
| ValidDER | cim:EnergySource.activePower |
| DERActivePowerInfeedDiffW | cim:SvPowerFlow.p |
| DERActivePowerInfeedDiffE | cim:SvPowerFlow.p |
| DERReactivePowerInfeedDiffW | cim:SvPowerFlow.q |
| ValidLoad | cim:EnergyConsumer.p |
| LoadActivePowerInfeedDiffW | cim:SvPowerFlow.p |
| LoadActivePowerInfeedDiffE | cim:SvPowerFlow.p |
| LoadReactivePowerInfeedDiffW | cim:SvPowerFlow.q |

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1673 **Table 17 (continuing) Restricted attribute values in an assembled model**

| Level 6 Rule | Restricted attribute value |
|---|---|
| ENIActivePowerInfeedLim | cim:ExternalNetworkInjection.p |
| ENIReactivePowerInfeedLim | cim:ExternalNetworkInjection.q |
| ENIActivePowerInfeedDiffW | cim:SvPowerFlow.p |
| ENIActivePowerInfeedDiffE | cim:SvPowerFlow.p |
| ENIReactivePowerInfeedDiffW | cim:SvPowerFlow.q |
| EIActivePowerInfeedLim | cim:EquivalentInjection.p |
| EIReactivePowerInfeedLim | cim:EquivalentInjection.q |
| EIActivePowerInfeedDiffW | cim:SvPowerFlow.p |
| EIActivePowerInfeedDiffE | cim:SvPowerFlow.p |
| EIReactivePowerInfeedDiffW | cim:SvPowerFlow.q |
| NetInterchange1 | cim:SvPowerFlow.p |
| NetInterchange2 | cim:SvPowerFlow.p |
| TapPosition | cim:SvTapStep.position |
| ShuntQ | cim:SvShuntCompensatorSections.sections |
| SvInjectionLimit | cim:SvInjection.pInjection,<br>cim:SvInjection.qInjection |
| VoltageProfile | cim:SvVoltage.v |
| VoltageTargetsAtTN | cim:RegulatingControl.targetValue |
| VoltageTargetAndDeadbandAtTN | cim:RegulatingControl.targetDeadband |
| EnergizedBoundaryTN | cim:SvVoltage.v |
| InvalidVoltage | cim:SvVoltage.v |
| DiscreteControl | cim:ShuntCompensator.sections,<br>cim:SvShuntCompensatorSections.sections,<br>cim:TapChanger.step, cim:SvTapStep.position |
| ContinousControl | cim:RegulatingControl.discrete,<br>cim:RegulatingControl.enabled |

1674

1675 The level 6 rules in Table 18 describe restrictions on number of instances for a class in an
1676 assembled model.

**Table 18 Restricted classes in an assembled model**

| Level 6 Rule | Restricted class |
|---|---|
| RequiredSvVoltage | cim:SvVoltage |
| RequiredSvSCSections | cim:SvShuntCompensatorSections |
| RequiredSvTapStep | cim:SvTapStep |

The level 7 rules in Table 19 describe restricted attribute values across different IGMs in an assembled model.

**Table 19 Restricted attribute values across IGMs in an assembled model**

| Level 7 Rule | Restricted attribute value |
|---|---|
| InconsistentCurrentLimits | cim:CurrentLimit.value |
| ACTielineBV | cim:BaseVoltage.nominalVoltage |
| HVDCScheduleMatch1 | cim:SvPowerFlow.p |
| HVDCScheduleMatch2 | cim:SvPowerFlow.p |
| ACScheduleMatch1 | cim:SvPowerFlow.p |
| ACScheduleMatch2 | cim:SvPowerFlow.p |
| NetInterchangeMatch1 | cim:ControlArea.netInterchange |
| NetInterchangeMatch2 | cim:ControlArea.netInterchange |

The level 7 rules in Table 20 describe restrictions on instances of classes across different IGMs in an assembled model.

**Table 20 Restricted relations across IGMs in an assembled model**

| Level 7 Rule | Restricted class |
|---|---|
| UnpairedTieFlow | cim:TopologicalNode (boundary), cim:EquivalentInjections, cim:ACLineSegment, cim:Switch |

The level 8 rules in Table 21 describe restrictions on the power flow calculation behaviour.



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1688 **Table 21 Rules on power flow behaviour**

| Level 8 Rule | Behaviour |
|---|---|
| CGMConvergence | cim:SvInjection.p, cim:SvInjection.q |
| IGMConvergence | cim:SvInjection.p, cim:SvInjection.q |
| CGMConvergenceRelaxed | cim:SvInjection.p, cim:SvInjection.q |

1689

1690 The level 8 rules in Table 22 describe restricted attribute values for a power flow solution.

1691 **Table 22 Restricted attribute values for a power flow solution.**

| Level 8 Rule | Restricted attribute value |
|---|---|
| CGMVoltageProfile | cim:SvVoltage.v |
| CGMCongestion | cim:CurrentLimit.value,<br>cim:ApparentPowerLimit.value |
| CGMTieFlowImbalance | cim:SvPowerFlow.p |

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